



सेंट्रल ट्रान्समिशन यटिलिटी ऑफ इंडिया लिमिटेड

(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)
(भारत सरकार का उद्यम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.

(A wholly owned subsidiary of Power Grid Corporation of India Limited)
(A Government of India Enterprise)

Ref: CTU/N/00/CMETS_NR/25

Date :15-11-2023

As per distribution list

Subject: 25th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

Dear Sir/Ma'am,

Please find enclosed the minutes of the 25th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 31st October, 2023 (Tuesday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,

(Handwritten signature)
15/11/23

(Kashish Bhambhani)
General Manager (CTU)

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Minutes of 25th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 31.10.23

The 25th Consultation Meeting for Evolving Transmission Schemes in Northern Region (CMETS-NR) was held through VC on 31/10/2023. List of participants is enclosed at **Annexure-1**

Confirmation of Minutes of 23rd & 24th CMETS NR

Regarding the minutes of 23rd & 24th CMETS NR meetings issued on 30/09/23 and 04/10/23 respectively, following observations were received from applicants/observed by CTUIL:

23rd CMETS-NR

- (i). It may be noted that for the Cannice Renewable Energy Pvt. Ltd. applications (0212100004: 80 MW, SW...1671600589106: 150 MW), 220kV S/c line was agreed for interconnection in ISTS however bay scope was inadvertently mentioned as 400 kV (1 No.) instead of 220 kV (1 no.). The same is to be considered as 220 kV(1 no.) and shall be considered in the intimations.
- (ii). It may be noted that the start date of connectivity for applicants at Bhadla-III PS granted with Bhadla-Fatehpur HVDC was mentioned as 30.06.2027(interim) considering the earlier approved timeline of 42 months. However, in the 15th NCT meeting held on 25.08.2023 (MoM issued on 18.09.2023), following was decided:

“After detailed deliberations, revised commissioning schedule of 48 months for Bipole-I and 54 months for Bipole-2 was recommended by NCT for the “Transmission system for evacuation of power from REZ in Rajasthan (20 GW) under Phase-III Part I”. Since this change in implementation timeline will have implication for evacuation from RE projects, recommendation will have to be sent to MoP in line with MoP order dated 22.03.23.”

In view of the above, the start date of connectivity for the respective cases shall be revised to 30.06.2028(considering implementation timeframe of 54 months) from earlier date of 30.06.2027(considering implementation timeframe of 42 months) in the intimations.

- (iii). In the minutes of 23rd CMETS NR meeting, for cases where connectivity is granted in sharing with other entities, the name of the entity which is sharing the bay granted to another entity (primary grantee of bay) is not mentioned in the connectivity system of the primary entity. The same shall be mentioned in the intimations of respective grantee(s) as both the cases were discussed and finalised in the same meeting.

- (iv). Subsequent to 23rd CMETS-NR meeting, representative of XL Xergi, Energizent & Teq Green vide mail dated 04.10.2023 had requested to include following in MOM.
- M/s XL Xergi requested that as GIB committee approval is yet to be applied/issued, therefore tower configuration shall be as per approval of GIB Committee.
 - M/s Energizent requested that as GIB committee approval is yet to be applied/issued, therefore tower configuration shall be as per approval of GIB Committee.
 - Connectivity of Teq Green (95 MW) is mentioned through sharing of Dedicated Transmission system of Energizent 220 kV S/c line. The same shall be S/c on D/c towers as the connectivity of Energizent is through S/c line of D/c towers.

The same were noted.

- (v). It may be noted that in case of Luceo Solar Pvt. Ltd. applications (0212100032: 200 MW & 0312100018:100 MW) the dedicated transmission system is inadvertently mentioned at Fatehgarh-IV (Section-2) PS, though in para same was mentioned as Fatehgarh-IV (Section-I) PS. The same is to be considered as Fatehgarh-IV (Section-I) PS and shall be considered in the intimations.
- (vi). It may be noted that in case of Connectivity of Helia Energy Pvt. Ltd. (App. No. 212100035:200 MW & 2200000120:100 MW), the dedicated transmission system is inadvertently mentioned as 220 kV D/c line. The same is to be considered as 220 kV S/c line on D/c tower and shall be considered in the intimations.
- (vii). It may be noted that for application of 50 MW by Sprng Pavana Urja(App No. 1200003719), the Start Date of Connectivity under GNA is inadvertently mentioned as 28.02.2026(Interim). The same is to be considered as 30.06.2027(interim) as per the applicant request in the meeting.
- (viii). It may be noted that in connectivity of Sprng Power Pvt. Ltd.(300 MW), Sprng Akshaya Urja(100 MW) & Sprng Energy Pvt. Ltd (400 MW) at Fatehgarh-III PS, the current carrying capacity of 400 kV dedicated line is inadvertently mentioned as “ (Suitable to carry minimum **800 MW** at nominal voltage)”. The same is to be considered as 900 MW and shall be considered in the intimations.

24th CMETS-NR

- (i). In the 24th CMETS NR meeting, implementation schedule of one set of Bus Sectionalizer at 220kV level of 400/220kV Fatehgarh-IV PS (Section-1) is mentioned as Feb'26 (Matching with Ph-IV (Part-2: 5.5 GW scheme). Subsequently, in the 15th CCTP meeting held on 12/10/23, project schedule was decided to be as 01st Feb'25 (Matching with Ph-III Part-A1 scheme)

Accordingly above respective Minutes were confirmed incorporating above observations.

1. Applications for Connectivity to ISTS under CERC (Connectivity and General Network Access to the inter-State Transmission System) Regulations, 2022 discussed in 23rd CMETS NR meeting:

In the 23rd CMETS-NR meeting, application for connectivity of 150 MW by M/s Solarcraft Power India 4 Pvt. Ltd. was discussed and proposed for grant at Bhadla-II PS in sharing basis with M/s Project Nine. Subsequently M/s Solarcraft vide mail dated 06.09.2023, informed that they wish to get connectivity at Bhadla-III PS instead of Bhadla-II PS as agreed in the meeting. Accordingly, it has been decided that the matter shall be taken up for discussion in the next CMETS-NR meeting. Therefore, above application is being discussed in the present meeting along with its other application received in month of Aug'23 as per respective application priority.

TABLE 1

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
1.	2200000152	Solarcraft Power India 4 Private Limited	Jaisalmer Rajasthan	18.07.2023	Generator (Solar)	Land Route	150	16.06.2025	Bhadla-III PS

2. Applications for Connectivity to ISTS under CERC (Connectivity and General Network Access to the inter-State Transmission System) Regulations, 2022 received in Aug'23:

Following 9 nos. of applications for Connectivity to ISTS are received in the month of Aug'23:

TABLE 2

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
1	2200000186	Renew Solar Power Private Limited	Barmer distt., Rajasthan	02.08.2023	Generator (Solar)	SECI LOA	300	30.06.2025	Fatehgarh-IV PS
2.	2200000187	Renew Solar Power Private Limited	Barmer distt., Rajasthan	02.08.2023	Generator (Solar)	SECI LOA	300	30.06.2025	Fatehgarh-IV PS
3.	2200000179	NTPC Renewable Energy Limited	Bikaner distt., Rajasthan	02.08.2023	Generator (Solar)	REMCL LOA	500	28.06.2025	Bikaner-III PS
4.	2200000181	Eden Renewable Cadet Private Limited	Barmer distt., Rajasthan	11.08.2023	Generator (Solar)	SECI LOA	300	26.04.2025	Fatehgarh-IV PS
5.	2200000184	Deshraj Solar Energy Private Limited	Bikaner distt., Rajasthan	11.08.2023	Generator (Solar)	Land Route	300	25.06.2025	Bikaner-III PS
6.	2200000185	Acme Heergarh Powertech Private Limited	Jodhpur distt., Rajasthan	14.08.2023	Generator (Solar)	Land Route	50	16.06.2025	Bhadla-II PS

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
7.	2200000192 (Enhancement)	Solarcraft Power India 4 Private Limited	Jaisalmer distt., Rajasthan	16.08.2023	Generator (Solar)	Land Route	110	16.06.2025	Bhadla-III PS
8.	2200000196	ACME Urja Two Private Limited	Chandauli, distt., Uttar Pradesh	23.08.2023	Standalone ESS (Hydro Pumped Storage)	--	600	01.01.2027	Varanasi S/s
9.	2200000212 (Enhancement)	Radiant Star Solar Park Private Limited	Barmer distt., Rajasthan	28.08.2023	Renewable Power Park Developer (Solar)	Land Route	100	31.12.2025	Fatehgarh-IV PS

1A) RE Applications for Connectivity to ISTS in July'23

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
1.	2200000152	Solarcraft Power India 4 Private Limited	Jaisalmer Rajasthan	18.07.2023	Generator (Solar)	Land Route	150	16.06.2025	Bhadla-III PS	Bhadla-III PS at 220 kV level	Conn-BG1: Rs. 50 lakh Conn-BG2: Nil (Bay under applicant scope) Conn-BG3: Rs.2 lakh/ MW

It was informed that in the 23rd CMETS-NR meeting, CTUIL proposed connectivity of M/s Solarcraft Power India 4 Pvt. Ltd. on sharing basis with M/s Project Nine Renewable granted connectivity at Bhadla-II PS. However, subsequently M/s Solarcraft vide mail dated 06.09.2023, informed that they wish to get connectivity at

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
Bhadla-III PS instead of Bhadla-II PS as agreed in the meeting. Accordingly, it was decided that the matter shall be taken up for discussion in the next CMETS-NR meeting. Subsequently, M/s Solarcraft Power India 4 Private Limited also applied for enhancement of connectivity for 110MW connectivity at Bhadla-III PS. In view of that, it was proposed to grant connectivity at Bhadla-III PS at 220kV level through a separate bay. M/s Solarcraft agreed for the same. M/s Solarcraft confirmed that the 220kV bay at Bhadla-III PS end shall be under applicant scope and the tower configuration shall be S/c on D/c tower. M/s Solarcraft was asked to submit necessary undertaking for tower configuration of S/c on D/c. M/s Solarcraft noted and agreed for the same. Further, it was also informed that M/s Solarcraft Power India 4 Private Limited has requested Start Date of Connectivity under GNA from 16/06/2025 for both the projects. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 30.06.2028 (interim). Further, it was also informed that application of connectivity of 50 MW by ACME at Bhadla-III PS was also agreed to be granted in sharing with M/s Solarcraft through same bay & DTL. M/s Solarcraft noted for the same. However, subsequent, to the meeting, M/s ACME vide letter dated 08.11.2023, informed their decision to withdraw their application of 50 MW, which was proposed in sharing. Accordingly, transmission system for connectivity of Solarcraft Power India 4 Private Limited under GNA Regulations 2022 shall be as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS): NIL</u> B. <u>Transmission System under applicant scope</u> (i). Common Pooling station of M/s Solarcraft Power India 4 Private Limited for Solar Power Projects with Appl. Nos.2200000152 (150 MW) & 2200000192 (110 MW) – Bhadla-III PS 220 kV S/c line on D/c towers (Suitable to carry minimum 300 MW at nominal voltage) C. <u>Transmission system for Connectivity under GNA (Augmentation without ATS)</u> 1. Augmentation of 1x500 MVA (7th) 400/220kV ICT at Bhadla-III PS 2. Augmentation of 2x1500 MVA (3rd & 4th), 765/400kV ICT at Bhadla-III PS 3. 765 kV Bhadla-III – Bikaner-III D/c line along with 240 MVar switchable line reactor for each circuit at Bhadla-III end 4. Establishment of 6000 MW, $\pm$ 800 kV Bhadla (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near Bhadla-III substation 5. Establishment of 6000 MW, $\pm$800 kV Fatehpur (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Fatehpur (UP) 6. Bhadla-III – Bhadla (HVDC) 400 kV D/c (1st & 2nd) quad moose line along with the line bays at both substations 7. Bhadla-III – Sikar-II 765 kV D/c line along with 330 MVar Switchable line reactor for each circuit at each end of Bhadla-III – Sikar-II 765 kV D/c line											

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
8. Fatehgarh-III – Bhadla-III 400kV D/c line (Quad) along with 63 MVAR Switchable line reactor for each circuit at both ends of Fatehgarh-III- Bhadla-III 400kV D/c line 9. Augmentation of 5x1500 MVA (1st to 5th), 765/400 kV ICTs at Fatehpur (HVDC) along with 2x330 MVAR (765kV) bus reactor 10. LILO of both ckts of 765 kV Varanasi – Kanpur (GIS) D/c line at Fatehpur 11. Augmentation by 1x1500 MVA ICT at 765/400kV Kanpur (GIS) substation											
Start Date of Connectivity under GNA: 30.06.2028(Interim) Final date shall be confirmed upon award of the system.											

1B) RE Applications for Connectivity to ISTS in Aug'23

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
1.	2200000186	Renew Solar Power Private Limited	Barmer distt., Rajasthan	02.08.2023	Generator (Solar)	SECI LOA	300	30.06.2025	Fatehgarh-IV PS	Fatehgarh-IV (Sec-2)	Conn-BG1:Rs. 50 lakh Conn-BG2: Rs. 6 Cr. Conn-BG3: Rs.2 lakh/MW
2.	2200000187	Renew Solar Power Private Limited	Barmer distt., Rajasthan	02.08.2023	Generator (Solar)	SECI LOA	300	30.06.2025	Fatehgarh-IV PS	Fatehgarh-IV (Sec-2)	Conn-BG1: Rs. 50 lakh Conn-BG2: NIL. Conn-BG3: Rs.2 lakh/ MW
It was informed that M/s Renew Solar Power Private Limited has filed two no. of connectivity applications of 300 MW each at Fatehgarh-IV PS. As informed in 23 rd CMETS-NR meeting, there is no margin available for grant of connectivity at 220 kV Fatehgarh-IV PS(Sec-2). Accordingly, considering the quantum of both applications (600 MW), it was proposed to grant connectivity at Fatehgarh-IV PS(Sec-2) at 400 kV level through 1 no. of 400 kV line bay. M/s Renew agreed for the same.M/s Renew informed that 400 kV bay at Fatehgarh-IV PS(Sec-II) end shall be under the scope of ISTS and the tower configuration shall be S/c on D/c tower. CTUIL asked M/s Renew to submit necessary undertaking for tower configuration of S/c on D/c. M/s Renew agreed for the same.											

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
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Further, it was also informed that M/s Renew Solar Power Private Limited has requested Start Date of Connectivity under GNA from 30/06/2025 for both applications. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 28.02.2026(interim). M/s Renew noted the same.

Transmission system for connectivity of M/s Renew Solar Power Private Limited under GNA Regulations 2022 is as below:

Details of Transmission system for Connectivity under GNA: (for both applications 2200000186(300 MW) & 2200000187 (300 MW))

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

- (i). Common Pooling station of M/s Renew Solar Power Private Limited Solar Power Projects (Appl. No. 2200000186 (300 MW) & Appl. No. 2200000187 (300 MW))– Fatehgarh-IV PS (Section-2) 400 kV S/c line on D/c towers (Suitable to carry minimum 900 MW per circuit at nominal voltage)

C. Common Transmission system for Connectivity under GNA

- 1) Augmentation of 2x1500 MVA(3rd & 4th), 765/400 kV ICT at Fatehgarh-IV (Section-2) Pooling Station
- 2) Fatehgarh-IV (Section-2) PS – Bhinmal (PG) 400 kV D/c line (Twin HTLS*) along with 50 MVAR switchable line reactor on each ckt at each end
- 3) LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line at Fatehgarh-IV (Section-2) PS along with 330 MVAR switchable line reactor at Fatehgarh-IV PS end of each ckt of 765 kV Fatehgarh-IV- Beawar D/c line (formed after LILO)
- 4) Establishment of 3x1500 MVA, 765/400 kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors & 2x125 MVAR, 420 kV Bus Reactor
- 5) Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end
- 6) Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVAR (765 kV) & 2x125 MVAR (400 kV) Bus Reactor
- 7) Fatehgarh-IV (Section-2) PS – Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end
- 8) Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVAR switchable line reactor for each circuit at Sirohi PS end (Quad)
- 9) Establishment of 2x1500 MVA, 765/400 kV Barmer-I Pooling Station along with 2x240 MVAR (765 kV) Bus Reactor & 2x125 MVAR, 420 kV Bus Reactor
- 10) Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad)
- 11) Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
12) Establishment of, 765 kV Substation at suitable location near Rishabdeo (Distt. Udaipur) along with 2x240 MVA (765 kV) Bus Reactor 13) Sirohi PS- Rishabdeo 765 kV D/c line along with 330 MVA switchable line reactor for each circuit at Sirohi end 14) Rishabdeo - Mandsaur PS 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at Rishabdeo end 15) LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo S/s 16) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVA switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line 17) Establishment of 765/400 kV (2x1500 MVA Kurawar S/s (with 1x500 MVA spare single phase transformer unit) with 2x330 MVA 765 kV bus reactor and 1x125 MVA 420 kV bus reactor (with 1x110 MVA & 1x80 MVA, 765 kV spare single phase reactor unit for line/bus reactor) 18) Mandsaur – Kurawar 765 kV D/c line along with 240 MVA switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 19) LILO of Indore – Bhopal 765 kV S/c line at Kurawar 20) Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 21) LILO of one circuit of Indore – Itarsi 400 kV D/c line at Ashtha 22) Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line * with minimum capacity of 2100 MVA/ckt at nominal voltage; Start Date of Connectivity under GNA: 28.02.2026(Interim). Final date shall be confirmed upon award of the system. It was also informed that with this grant of 600 MW to M/s Renew, the total connectivity granted at Fatehgarh-IV PS(Sec-II) is 4030 MW. As the transmission system for Rajasthan Ph-IV Part-2(5.5 GW) was planned for 4 GW at Fatehgarh-IV PS(Sec-2) & Barmer-I PS(1.5 GW), for further grant of connectivity at 400 kV Fatehgarh-IV PS(Sec-II), for additional requirement, additional transmission system beyond Fatehgarh-IV PS(Sec-II) need to be evolved.											
3.	2200000179	NTPC Renewable Energy Limited	Bikaner distt., Rajasthan	02.08.2023	Generator (Solar)	REMCL LOA	500	28.06.2025	Bikaner-III PS	Bikaner-IV PS at 400 kV level	Conn-BG1: Rs. 50 lakh Conn-BG2: Rs. 6 Cr. Conn-BG3: Rs.2 lakh/ MW
It was informed that M/s NTPC Renewable Energy Limited had applied for connectivity for 500 MW at Bikaner-III PS. As informed in 23rd CMETS-NR meeting, there is no margin available for grant of connectivity at Bikaner-III PS(except 83 MW margin in sharing). Further another application of 150MW was received in the month of Sep'23. Accordingly, it was proposed to grant connectivity to NTPC for cumulative quantum of 650MW (500MW+150MW) at Bikaner-IV PS at 400 kV level through 1 no. of 400 kV line bay. NTPC agreed for the same. It was also informed that Transmission scheme for Bikaner-IV PS is being taken up for discussion and approval in the present meeting.											

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M/s NTPC informed that 400 kV bay at Bikaner-IV PS end shall be under the scope of ISTS and the tower configuration shall be S/c on D/c tower. CTUIL asked M/s NTPC to submit necessary undertaking for tower configuration of S/c on D/c. M/s NTPC agreed for the same. It was also informed that M/s NTPC has requested Start Date of Connectivity under GNA from 28/06/2025. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 30.06.2026 (interim). M/s NTPC noted the same. Transmission system for connectivity of NTPC Renewable Energy Limited under GNA Regulations 2022 is as below: <u>Details of Transmission system for Connectivity under GNA: (500 MW)</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Common pooling station for NTPC Renewable Energy Limited Solar Power Projects (App No.2200000179(500 MW) & 2200000244(150 MW) – Bikaner-IV PS 400 kV S/c line on D/c towers (Suitable to carry minimum 900 MW at nominal voltage) C. <u>Common Transmission system for Connectivity under GNA</u> 1) Establishment of 2x1500 MVA, 765/400 kV Bikaner-IV Pooling Station along with 2x240 MVAR (765kV) Bus Reactor & 2x125 MVAR (420kV) Bus Reactor at a suitable location near Bikaner 2) LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line (2nd) at Bikaner-IV PS along with 330 MVAR switchable line reactor at Bikaner-IV PS end of each ckt of 765 kV Bikaner-IV -Neemrana-II line (formed after LILO) 3) Establishment of 765/400 kV, 2x1500 MVA S/s at suitable location near Churu along with 2x240 MVAR (765kV) Bus Reactor & 2x125 MVAR (420kV) Bus Reactor 4) Bikaner-IV PS – Churu 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at Bikaner-IV PS end 5) LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s 6) Churu – Fatehabad (PG) 400 kV D/c line (Quad Moose) along with 63 MVAR switchable line reactor for each circuit at Churu S/s end 7) Establishment of 765/400 kV, 3x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVAR (765kV) Bus Reactor & 2x125 MVAR (420kV) Bus Reactor 8) Siwani – Jind (PG) 400 kV D/c line (Quad Moose)											

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9) Siwani – Patran (Indi Grid) 400 kV D/c line (Quad Moose) along with 80 MVAR switchable line reactor for each circuit at Siwani S/s end 10) STATCOM (2x+300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR) along with 2 nos. 400kV line bays at Bikaner-IV PS D. Additional Transmission system for Inter-regional power transfer: 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 30.06.2026(Interim). Final date shall be confirmed upon award of the system											
4.	2200000181	Eden Renewable Cadet Private Limited	Barmer distt., Rajasthan	11.08.2023	Generator (Solar)	SECI LOA	300	26.04.2025	Fatehgarh-IV PS	Barmer-I PS at 220 kV level	Conn-BG1: Rs. 50 lakh Conn-BG2: Rs. 3 Cr. Conn-BG3: Rs.2 lakh/ MW
It was informed that M/s Eden Renewable cadet had applied for connectivity for 300 MW at Fatehgarh-IV PS. However, as informed in 23rd CMETS-NR meeting, there is no margin available for grant of connectivity at 220 kV Fatehgarh-IV PS(Sec-2). Accordingly, it was proposed to grant connectivity at Barmer-I PS at 220 kV level through 1 no. of 220 kV line bay. Applicant may confirm the 220kV bay scope at Barmer-I PS end and the tower configuration. M/s Eden informed that 220 kV bay at Barmer-I PS end shall be under the scope of ISTS and the tower configuration shall be S/c on D/c tower. CTUIL asked M/s Eden to submit necessary undertaking for tower configuration of S/c on D/c. M/s Eden agreed for the same. M/s Eden stated that they have applied for SECI bid considering other locations than Barmer-I PS. CTU stated that applications are being processed on first come first serve basis and 220kV Fatehgarh-IV (Sec-II) is already closed for any further grant. Considering above, Barmer location is proposed for connectivity. Eden requested that they will revert on the matter after discussing with SECI in 1-2 days. CTU stated that in case of no response in given timelines, CTU will proceed with proposed connectivity only. It was also informed that M/s Eden has requested Start Date of Connectivity under GNA from 26/04/2025. However, considering the expected commissioning date											

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of transmission system, the start date of Connectivity under GNA shall be 28.02.2026(interim). M/s Eden noted the same. Transmission system for connectivity of M/s Eden Renewable Cadet under GNA Regulations 2022 is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Eden Renewable Cadet Private Limited – Barmer-I PS 220 kV S/c line on D/c towers (Suitable to carry minimum 300 MW at nominal voltage) C. <u>Common Transmission system for Connectivity under GNA</u> 1) 1x500 MVA (2nd), 400/220 kV ICT at Barmer-I Pooling Station 2) 2x1500 MVA (2nd & 3rd), 765/400 kV ICTs at Barmer-I Pooling Station 3) Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad) 4) Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVar (765 kV) & 2x125 MVar (400 kV) Bus Reactor 5) Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 6) Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVar switchable line reactor for each circuit at Sirohi PS end (Quad) 7) Establishment of, 765 kV Substation at suitable location near Rishabdeo (Distt. Udaipur) along with 2x240 MVar (765 kV) Bus Reactor 8) Sirohi PS- Rishabdeo 765 kV D/c line along with 330 MVar switchable line reactor for each circuit at Sirohi end 9) Rishabdeo - Mandsaur PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at Rishabdeo end 10) LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo S/s 11) Establishment of 765 kV Mandsaur PS along with 2x330 MVAR (765 kV) Bus Reactors 12) Mandsaur PS–Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line 13) Establishment of 765/400 kV (2x1500 MVA Kurawar S/s (with 1x500 MVA spare single phase transformer unit) with 2x330 MVar 765 kV bus reactor and 1x125 MVar 420 kV bus reactor (with 1x110 MVar & 1x80 MVar, 765 kV spare single phase reactor unit for line/bus reactor) 14) Mandsaur – Kurawar 765 kV D/c line along with 240 MVar switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 15) LILO of Indore – Bhopal 765 kV S/c line at Kurawar											

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16) Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 17) LILO of one circuit of Indore – Itarsi 400 kV D/c line at Astha 18) Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line											
Start Date of Connectivity under GNA: 28.02.2026(Interim). Final date shall be confirmed upon award of the system											
5.	2200000184	Deshraj Solar Energy Private Limited	Bikaner distt., Rajasthan	11.08.2023	Generator (Solar)	Land Route	300	25.06.2025	Bikaner-III PS	Bikaner-IV PS at 220 kV level	Conn-BG1: Rs. 50 lakh Conn-BG2: Rs. 3 Cr. Conn-BG3: Rs.2 lakh/ MW
It was informed that M/s Deshraj Solar Energy Private Limited had applied for connectivity for 300 MW at Bikaner-III PS. However, as informed in 23rd CMETS-NR meeting, there is no margin available for grant of connectivity at Bikaner-III PS(except 83 MW margin in sharing). Accordingly, it is proposed to grant connectivity at Bikaner-IV PS at 220 kV level through 1 no. of 220 kV line bay. It was also informed that Transmission scheme for Bikaner-IV PS is being taken up for discussion and approval in the present meeting. M/s Deshraj informed that 220 kV bay at Bikaner-IV PS end shall be under the scope of ISTS and the tower configuration shall be S/c on D/c tower. CTUIL asked M/s Deshraj to submit necessary undertaking for tower configuration of S/c on D/c. M/s Deshraj agreed for the same. M/s Deshraj Solar enquired about impact of further connectivity at Bikaner-III with provision of BESS capacity. CTU clarified that even RE with BESS cannot be provided on Bikaner-III as it would lead to additional injection which is not feasible. However in case of standalone BESS, it will depend on the capacities, location and future bids coming out for which timeframe. M/s Deshraj noted the same. It was also Informed that M/s Deshraj has requested Start Date of Connectivity under GNA from 25/06/2025. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 30.06.2026 (interim). M/s Deshraj noted the same. Transmission system for connectivity of Deshraj Solar Energy Private Limited under GNA Regulations 2022 is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u>											

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i. Deshraj Solar Energy Private Limited – Bikaner-IV PS 220 kV S/c line on D/c towers (Suitable to carry minimum 300 MW at nominal voltage) C. <u>Common Transmission system for Connectivity under GNA</u> - Establishment of 2x1500 MVA, 765/400 kV & 1x500MVA 400/220kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor at a suitable location near Bikaner - LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line (2nd) at Bikaner-IV PS along with 330 MVAR switchable line reactor at Bikaner-IV PS end of each ckt of 765 kV Bikaner-IV -Neemrana-II line (formed after LILO) - Establishment of 765/400 kV, 2x1500 MVA S/s at suitable location near Churu along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor - Bikaner-IV PS – Churu 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at Bikaner-IV PS end - LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s - Churu – Fatehabad (PG) 400 kV D/c line (Quad Moose) along with 63 MVar switchable line reactor for each circuit at Churu S/s end - Establishment of 765/400 kV, 3x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor - Siwani – Jind (PG) 400 kV D/c line (Quad Moose) - Siwani – Patran (Indi Grid) 400 kV D/c line (Quad Moose) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end - STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV line bays at Bikaner-IV PS D. <u>Additional Transmission system for Inter-regional power transfer:</u> - Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors - Beawar- Mandsaur PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVar 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVar switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 30.06.2026(Interim). Final date shall be confirmed upon award of the system											

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6.	2200000185	Acme Heergarh Powertech Private Limited	Jodhpur distt., Rajasthan	14.08.2023	Generator (Solar)	Land Route	50	16.06.2025	Bhadla-II PS	Bhadla-III PS	Conn-BG1: Rs. 50 lakh Conn-BG2: Rs. NIL Conn-BG3: Rs.2 lakh/ MW

It was informed that M/s Acme Heergarh Powertech Private Limited has applied for connectivity for 50 MW at Bhadla-II PS. However, it was informed that subsequent to the 23rd CMETS-NR meeting held on 29/08/23, M/s Eden filed petitions with CERC for connectivity at Fatehgarh-III PS & Bhadla-II PS and the matter is currently in adjudication in CERC. Further, CERC during hearing on 25.10.2023 (RoP dated 31.10.23) CTUIL to maintain the status quo with regard to the grant of connectivity/allocating the bays at Bhadla II PS. M/s ACME enquired about grant of connectivity at 220 kV Bhadla-II(Sec-I) in sharing with existing project of M/s ACME. Regarding this, CTUIL informed that Bhadla-II S/s is already closed for fresh grant of connectivity except for the capacity of surrendered bays (220 kV 2 no. i.e. 600 MW) regarding which matter is with CERC. Therefore, there is no margin available for grant of additional connectivity at Bhadla-II PS.

In view of above, connectivity to M/s ACME was proposed at Bhadla-III PS at 220 kV level in sharing with M/s Solarcraft Power India 4 Private Limited instead of Bhadla-II.M/s ACME agreed for the same.

Further, it was also informed that M/s Acme Heergarh Powertech Private Limited has requested Start Date of Connectivity under GNA from 16/06/2025. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 30.06.2028 (interim). M/s ACME noted the same.

Transmission system for connectivity of M/s Acme Heergarh Powertech Private Limited under GNA is as below:

A. Details of Transmission system for Connectivity under GNA:

B. Associated Transmission System (ATS): NIL

Transmission System under applicant scope

- (i). Through sharing of dedicated transmission system of M/s Solarcraft Power India 4 Private Limited for Solar Power Projects with Appl. Nos.2200000152 (150 MW) & 2200000192 (110 MW) – Bhadla-III PS 220 kV S/c line on D/c towers (Suitable to carry minimum 310 MW at nominal voltage)

C. Common Transmission system for Connectivity under GNA

1. Augmentation of 1x500 MVA (7th) 400/220kV ICT at Bhadla-III PS

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
2. Augmentation of 2x1500 MVA (3rd & 4th), 765/400kV ICT at Bhadla-III PS 3. 765 kV Bhadla-III – Bikaner-III D/c line along with 240 MVAR switchable line reactor for each circuit at Bhadla-III end 4. Establishment of 6000 MW, $\pm$ 800 kV Bhadla (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near Bhadla-III substation 5. Establishment of 6000 MW, $\pm$ 800 kV Fatehpur (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Fatehpur (UP) 6. Bhadla-III – Bhadla (HVDC) 400 kV D/c (1st & 2nd) quad moose line along with the line bays at both substations 7. Bhadla-III – Sikar-II 765 kV D/c line along with 330 MVAR Switchable line reactor for each circuit at each end of Bhadla-III – Sikar-II 765 kV D/c line 8. Fatehgarh-III – Bhadla-III 400kV D/c line (Quad) along with 63 MVAR Switchable line reactor for each circuit at both ends of Fatehgarh-III- Bhadla-III 400kV D/c line 9. Augmentation of 5x1500 MVA (1st to 5th), 765/400 kV ICTs at Fatehpur (HVDC) along with 2x330 MVAR (765kV) bus reactor 10. LILO of both ckts of 765 kV Varanasi – Kanpur (GIS) D/c line at Fatehpur 11. Augmentation by 1x1500 MVA ICT at 765/400kV Kanpur (GIS) substation Start Date of Connectivity under GNA: 30.06.2028(Interim) Final date shall be confirmed upon award of the system. However, subsequent to the meeting, M/s ACME vide letter dated 08.11.2023, informed their decision to withdraw their application of 50 MW. Accordingly, the application shall be closed and treatment shall be as per GNA Regulations.											
7.	2200000192 (Enhancement)	Solarcraft Power India 4 Private Limited	Jaisalmer distt., Rajasthan	16.08.2023	Generator (Solar)	Land Route	110	16.06.2025	Bhadla-III PS	Bhadla-III PS	Conn-BG1: Rs. 50 lakh Conn-BG2: Rs. NIL Conn-BG3: Rs.2 lakh/ MW
As discussed under application no 2200000152(150 MW) of M/s Solarcraft, it was proposed to grant connectivity of 110 MW at Bhadla-III PS at 220kV level in sharing with other application of M/s Solarcraft(150 MW). M/s Solarcraft agreed for the same. It was also informed that connectivity of 50 MW to M/s ACME Heergarh Pvt. Ltd is also granted in sharing with M/s Solarcraft through same bay & DTL. Further, it was informed that M/s Solarcraft Power India 4 Private Limited has requested Start Date of Connectivity under GNA from 16/06/2025. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 30.06.2028 (interim). M/s Solarcraft noted the same. However, subsequent, to the meeting, M/s ACME vide letter dated 08.11.2023, informed their decision to withdraw their application of 50 MW, which was proposed in sharing. Transmission system for connectivity of Solarcraft Power India 4 Private Limited under GNA is as below:											

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
Details of Transmission system for Connectivity under GNA:											
A. <u>Associated Transmission System (ATS):</u> NIL											
B. <u>Transmission System under applicant scope</u>											
(i). Common Pooling station of M/s Solarcraft Power India 4 Private Limited for Solar Power Projects with Appl. Nos.2200000152 (150 MW) & 2200000192 (110 MW) – Bhadla-III PS 220 kV S/c line on D/c towers(Suitable to carry minimum 310 MW at nominal voltage)											
C. <u>Common Transmission system for Connectivity under GNA</u>											
1. Augmentation of 2x500 MVA (7 th & 8 th) 400/220kV ICTs at Bhadla-III PS											
2. Augmentation of 2x1500 MVA (3rd & 4th), 765/400kV ICT at Bhadla-III PS											
3. 765 kV Bhadla-III – Bikaner-III D/c line along with 240 MVar switchable line reactor for each circuit at Bhadla-III end											
4. Establishment of 6000 MW, ± 800 kV Bhadla (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near Bhadla-III substation											
5. Establishment of 6000 MW, ±800 kV Fatehpur (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Fatehpur (UP)											
6. Bhadla-III – Bhadla (HVDC) 400 kV D/c (1st & 2nd) quad moose line along with the line bays at both substations											
7. Bhadla-III – Sikar-II 765 kV D/c line along with 330 MVar Switchable line reactor for each circuit at each end of Bhadla-III – Sikar-II 765 kV D/c line											
8. Fatehgarh-III – Bhadla-III 400kV D/c line (Quad) along with 63 MVar Switchable line reactor for each circuit at both ends of Fatehgarh-III- Bhadla-III 400kV D/c line											
9. Augmentation of 5x1500 MVA (1st to 5th), 765/400 kV ICTs at Fatehpur (HVDC) along with 2x330 MVar (765kV) bus reactor											
10. LILO of both ckts of 765 kV Varanasi – Kanpur (GIS) D/c line at Fatehpur											
11. Augmentation by 1x1500 MVA ICT at 765/400kV Kanpur (GIS) substation											
Start Date of Connectivity under GNA: 30.06.2028(Interim) Final date shall be confirmed upon award of the system.											
8.	2200000212 (Enhancement)	Radiant Star Solar Park Private Limited	Barmer distt., Rajasthan	28.08.2023	Renewable Power Park Developer (Solar)	Land Route	100	31.12.2025	Fatehgarh-IV PS	Barmer-I PS (at 220 kV)	Conn-BG1: Rs. 50 lakh Conn-BG2: Rs. 3 Cr. Conn-BG3: Rs.2 lakh/ MW

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
It was informed that M/s Radiant Star Solar Park Private Limited had been earlier granted connectivity of 200 MW at Fatehgarh-IV PS(Sec-II) at 220 kV(1 no.). In the 23rd CMETS NR meeting, connectivity of 50 MW to M/s Avaada was granted in sharing with M/s Radiant. Further, as informed in 23rd CMETS meeting. there is no additional margin available at 220 kV Fatehgarh-IV PS(Sec-II) except of 50 MW at 220 kV bay granted to M/s Radiant. Subsequently, M/s Radiant has applied for enhancement of connectivity for 100 MW at Fatehgarh-IV PS(Sec-II). In view of the above, it was proposed to grant connectivity at Barmer-I PS. M/s Radiant informed that 300 MW capacity is of same solarpark for which they have applied for two no. of applications of 200 & 100 MW. It will not be viable to connect same project to different substations through. M/s Radiant also enquired about the status of M/s Avaada whether they are still going ahead with their 50 MW at Fatehgarh-IV PS in sharing. Regarding this CTUIL informed that , as on date there is no information from M/s Avaada on not going ahead in this regard. However, in case of any vacancy created at Fatehgarh—IV PS (Sec-II), the same shall be offered to applicants granted connectivity at Barmer-I PS as per the application priority. M/s Radiant noted the same. It was informed that other connectivity grantees at Barmer-I PS at 220 kV are of 300-400 MW and therefore no capacity margin is available in those 220 kV bays for granting connectivity of 100 MW in sharing. In view of the above, though M/s Radiant has applied for only 100MW,it was proposed to grant connectivity at Barmer-I PS at 220 kV level through 1 no. of 220 kV line bay. Balance margin (200 MW) will be available in 220 kV bay, which shall be granted to new applicants in sharing basis with M/s Radiant who agreed for the same. Subsequently, M/s Radiant vide mail dated 02/11/2023 confirmed their acceptance of connectivity at Barmer-I PS for 100 MW. M/s Radiant informed that 220 kV bay at Barmer-I PS end shall be under the scope of ISTS and the tower configuration shall be S/c on D/c tower. CTUIL asked M/s Radiant to submit necessary undertaking for tower configuration of S/c on D/c. M/s Radiant agreed for the same. Further, it was also informed that M/s Radiant Star Solar Park Private Limited has requested Start Date of Connectivity under GNA from 31/12/2025. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 28.02.2026(interim). M/s Radiant noted the same. Transmission system for connectivity of M/s Radiant Star Solar Park under GNA is as below: <u>Details of Transmission system for Connectivity under GNA: (100 MW)</u> A. <u>Associated Transmission System (ATS): NIL</u> B. <u>Transmission System under applicant scope</u> (i). Radiant Star Solar Park Private Limited – Barmer-I PS 220 kV S/c line on D/c towers (Suitable to carry minimum 300 MW at nominal voltage)											

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
C. Common Transmission system for Connectivity under GNA											
1) 2x1500 MVA (2 nd & 3 rd), 765/400 kV ICTs at Barmer-I Pooling Station 2) 1x500 MVA (2 nd), 400/220 kV ICT at Barmer-I Pooling Station 3) Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad) 4) Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVar (765 kV) & 2x125 MVar (400 kV) Bus Reactor 5) Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 6) Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVar switchable line reactor for each circuit at Sirohi PS end (Quad) 7) Establishment of, 765 kV Substation at suitable location near Rishabdeo (Distt. Udaipur) along with 2x240 MVar (765 kV) Bus Reactor 8) Sirohi PS- Rishabdeo 765 kV D/c line along with 330 MVar switchable line reactor for each circuit at Sirohi end 9) Rishabdeo - Mandsaur PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at Rishabdeo end 10) LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo S/s 11) Establishment of 765 kV Mandsaur PS along with 2x330 MVAR (765 kV) Bus Reactors 12) Mandsaur PS–Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line 13) Establishment of 765/400 kV (2x1500 MVA Kurawar S/s (with 1x500 MVA spare single phase transformer unit) with 2x330 MVar 765 kV bus reactor and 1x125 MVar 420 kV bus reactor (with 1x110 MVar & 1x80 MVar, 765 kV spare single phase reactor unit for line/bus reactor) 14) Mandsaur – Kurawar 765 kV D/c line along with 240 MVar switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 15) LILO of Indore – Bhopal 765 kV S/c line at Kurawar 16) Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line 17) LILO of one circuit of Indore – Itarsi 400 kV D/c line at Ashtha 18) Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line											
Start Date of Connectivity under GNA: 28.02.2026(Interim). Final date shall be confirmed upon award of the system											

1B) Conventional Applications received in month of Aug'23 for Connectivity to ISTS

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
1.	2200000196	ACME Urja Two Private Limited	Chandauli, distt., Uttar Pradesh	23.08.2023	Standalone ESS (Hydro Pumped Storage)	LOA or PPA	600	01.01.2027	Varanasi S/s	Varanasi S/s at 400 kV	Conn-BG1: Rs. 50 lakh Conn-BG2: Rs. 12 Cr Conn-BG3: Rs.2 lakh/MW

It was informed that M/s ACME Urja Two Private Limited has applied for connectivity of 600MW for its Standalone ESS (Hydro Pumped Storage) project for connectivity at 400kV Varanasi S/s. As per the application, applicant applied for maximum injection of 600MW (Injection cycle: 10 hours) and maximum drawl of 600MW (drawl cycle: 8 hours).

In the CEA meeting held on 16.10.23, applicability of N-1 criteria to the immediate connectivity system of Pump storage Projects with Grid was deliberated and it was decided that N-1 criterion would be applicable on to the immediate connectivity system for Pump storage Projects.

765/400kV Varanasi (PG) substation is existing substation and is about 70 kms from project locations as per the coordinates provided in the application. POWERGRID vide mail 18.10.23 confirmed the availability of 2 nos. of 400kV line bays at Varanasi substation. From the studies it is emerged that when pumped storage plant works in motoring mode (Drawl of 600MW power), loading on 765/400kV ICTs (2x1500MVA) at Varanasi is about 1400MW in N-1 condition of solar maximized scenario in 2027-28 timeframe considering Bhadla-Fatehpur HVDC system, which is in order. In other scenarios (with and without HVDC system), no constraint is observed beyond Varanasi Substation for injection or drawl of power.

On query of CEA, M/s ACME clarified that they are yet to submit DPR for above project to CEA. However, project DPR is already prepared, water allocations is also done along with consent of UP govt received. M/s ACME stated that are planning additional 1200 MW PSP in same complex, however connectivity application for same is not yet filed. Regarding applicant query on scope of bays at ISTS substation end, CTU stated that bays at ISTS end can be either implemented in ISTS scope or developer scope. However in case of hydro projects, it is recommended that developer may implement bays in their scope matching with generation project. In case bay(s) are being developed in ISTS, Conn BG-2 to be submitted by applicant as per GNA regulation and in case of generation

delay, transmission charges shall be applicable as per CERC regulations. Applicant informed that they will inform about bay scope at ISTS substation end within 1-2 days.

Further M/s ACME Urja Two Private Limited vide letter dated 01.11.23 informed that scope for 2 nos. of 400kV bays will be in ISTS scope and they will submit the requisite BGs as per GNA regulations. Applicant also informed that they would connect their hydro PSP to Varanasi (PG) S/s through D/c line on D/c tower.

M/s ACME stated that they are Planning to apply for additional 1200MW PSP for injection at Varanasi S/s in 2-3 months. CTU stated that they may interconnect future hydro projects through same dedicated transmission system based on capacity of dedicated transmission system which at present is planned commensurate to 600 MW requirement, however, there will be additional system requirement which will be identified at later stage after receipt of additional connectivity application.

In view of the above, it was agreed to grant connectivity to M/s ACME Urja Two Private Limited with following connectivity system.

Details of Transmission system for Connectivity under GNA (600MW):

A. Transmission System under applicant scope

- M/s ACME Urja Two Private Limited – Varanasi 400kV D/c line along with associated bays at generation (PSP) end (Suitable to carry minimum 900 MW/ckt at nominal voltage)- about 70 kms

B. Existing system beyond 400 kV Varanasi S/s

3. Applications for Connectivity to ISTS under CERC (Connectivity and General Network Access to the inter-Stat Transmission System) Regulations, 2022 received in Sep'23:

Following 8 Nos. of applications for Connectivity to ISTS are received in the month of Sep'23

TABLE 2

Sl. No.	Applicati on ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
1.	2200000227	Sunsure Solarpark Rj One Private Limited	Bikaner distt., Rajasthan	01.09.2023	Generator (Solar)	Land BG Route	50	31.03.2026	Bikaner-III PS
2.	2200000228	Sunsure Solarpark Rj One Private Limited	Bikaner distt., Rajasthan	01.09.2023	Generator (Solar)	Land BG Route	50	31.03.2026	Bikaner-III PS
3.	2200000210	Solarcraft Power India 17 Private Limited	Bikaner distt., Rajasthan	05.09.2023	Generator (Solar)	Land Route	300	30.12.2025	Bikaner-III PS
4.	2200000244	NTPC Renewable Energy Limited	Bikaner distt., Rajasthan	07.09.2023	Generator (Solar)	LOA or PPA	150	28.06.2025	Bikaner-III PS
5.	2200000232	Green Infra Clean Energy Generation Limited	Jaisalmer distt., Rajasthan	08.09.2023	Generator (Solar)	Land Route	300	01.06.2025	Bhadla-III PS
6.	2200000221	First Energy Private Limited	Bikaner distt., Rajasthan	11.09.2023	Captive Generating Plant (Solar)	Land BG Route	100	31.05.2025	Bikaner-II PS
7.	2200000285	Sunsure Solarpark Fourteen Pvt. Limited	Bikaner distt., Rajasthan	30.09.2023	Generator (Solar)	Land BG Route	300	31.03.2027	Bikaner-III PS
8.	2200000286	Enren-I Energy Private Limited	Barmer distt., Rajasthan	30.09.2023	Generator (Solar)	Land BG Route	300	25.02.2026	Barmer-I PS

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
1.	2200000227	Sunsure Solarpark Rj One Private Limited	Bikaner distt., Rajasthan	01.09.2023	Generator (Solar)	Land BG Route	50	31.03.2026	Bikaner-III PS	Bikaner-III PS	Conn-BG1:Rs. 50 lakh Conn-BG2: NIL Conn-BG3: Rs.2 lakh/MW

It was informed that M/s Sunsare Solarpark RJ One Private Limited has applied for connectivity for 50 MW at Bikaner-III PS. However, as informed in 23rd CMETS-NR meeting, Bikaner-III PS was planned for 4000 MW and presently 3917 MW of connectivity is granted at Bikaner-III PS. There is a margin of 83 MW available for grant of connectivity at 220 kV Bikaner-III PS (including the 33 MW margin available at 220 kV bay granted to M/s Proteus(217 MW) which is shared with Avaada(50 MW)). It was informed that Sunsare Solarpark Rj One Private Limited is already granted connectivity of 150 MW at Bikaner-III PS at 220 kV bay in sharing with Saimaa Solar(150 MW). Accordingly, it was proposed to grant present application of 50 MW in sharing with earlier applications of Sunsare(150 MW) and Saimaa(150 MW) taking total connectivity of 200 MW to Sunsare(350 MW including Saimaa). M/s Sunsare agreed for the same.

It was also informed that M/s Sunsare Solarpark RJ One Private Limited has requested Start Date of Connectivity under GNA from 31/03/2026. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 31.03.2026(interim). M/s Sunsare noted the same.

Transmission system for connectivity of M/s Sunsare Solarpark Rj One Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

- (i). Common pooling station for Sunsare Solarpark RJ One Private Limited Solar Projects(App. No. 2200000172(50 MW), 2200000165(50 MW), 2200000164(50 MW) & 2200000227(50 MW) + Saimaa Solar (App No. 2200000167 (150 MW))– Bikaner-III PS 220 kV S/c line on D/c towers(Suitable to carry minimum 350 MW at nominal voltage)

C. Common Transmission system for Connectivity under GNA:

1. Augmentation of 1x500 MVA (4th) 400/220 kV ICT at Bikaner-III Pooling Station

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
2. Augmentation of 2x1500 MVA (5th & 6th) 765/400 kV ICTs at Bikaner-III Pooling Station 3. Augmentation of 1x1500 MVA (4th) 765/400kV ICT at Bikaner (PG) 4. LILO of both ckts of 400 kV Bikaner(PG)-Bikaner-II D/c line (Quad) at Bikaner-III PS 5. Bikaner-II PS – Bikaner-III PS 400 kV D/c line (Quad) 6. Bikaner-III - Neemrana-II 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at each end 7. Augmentation of 2x1500 MVA (3rd & 4th), 765/400 kV ICTs at Neemrana-II S/s 8. Neemrana-II -Kotputli 400 kV D/c line (Quad) 9. LILO of both ckts of 400 kV Gurgaon (PG) - Sohna Road (GPTL) D/c line (Quad) at Neemrana-II S/s 10. Bikaner-III - Neemrana-II 765 kV D/c line (2nd) along with 330 MVAR switchable line reactor for each circuit at each end 11. Neemrana-II- Bareilly (PG) 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at each end 12. Augmentation by 400/220 kV, 1x500 MVA ICT (3rd) at Kotputli (PG) 13. 400 kV Bareilly(765/400kV) – Bareilly(PG) D/c line (Quad) (2nd) 14. Augmentation with 1x1500 MVA, 765/400 kV ICT at Bareilly (765/400kV) S/s (3rd)											
D. Additional Transmission system for Inter-regional power transfer: 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar											
Start Date of Connectivity under GNA: 31.03.2026(Interim). Final date shall be confirmed upon award of the system.											
2.	2200000228	Sunsure Solarpark Rj One Private Limited	Bikaner distt., Rajasthan	01.09.2023	Generator (Solar)	Land BG Route	50	31.03.2026	Bikaner-III PS	Bikaner-III PS	Conn-BG1: Rs. 50 lakh Conn-BG2: NIL Conn-BG3: Rs.2 lakh/ MW

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
It was informed that M/s Sunsire Solarpark Rj One Private Limited has additionally applied for connectivity for 50 MW at Bikaner-III PS. After grant of 50 MW to Sunsire for application no. (2200000227), there is balance margin of only 33 MW available for grant of connectivity at Bikaner-III PS. Considering requirement of additional 17 MW connectivity (50-33 MW), based on margin availability, same can be granted at Bikaner-III PS, however this can be accommodated on the bay having margin i.e. Proteus. Further, it was informed that no additional connectivity including enhancement will be available henceforth on Bikaner-III PS. In view of the above, it was proposed to grant connectivity of 50 MW in sharing with M/s Proteus (217 MW) & Avaada (50 MW) which is granted connectivity at 220 kV Bikaner-III PS. M/s Sunsire reiterated that as per their mail dated 30.10.23, that allocation of connectivity at different bays shall increase their project overall cost and maintenance cost over the lifetime of the project which will impact the viability of the complete project. Therefore M/s Sunsire requested to review sharing of connectivity among different entities at Bikaner-III PS so that they can be accommodated 250 MW(all 5 applications of Sunsire of 50 MW each) at a single bay. M/s Sunsire vide mail dated 30.10.2023 informed that one possible option is Bay no. 2 can be allocated to Proteus Energy Private Limited (217MW) & Saimaa Solar Private Limited (150MW). Accordingly, Bay no. 5 can be allocated to Sunsire Solarpark RJ One Private Limited (250MW) & Avaada RJ Bikaner Private Limited (50MW). By choosing this option, both bays will be completely utilised and allow us to construct our solar project at a single location._ CTUIL informed that for discussion of matter related to sharing of bays, all other entities granted connectivity at Bikaner-II PS are to be invited. As other entities are not available in the meeting, a separate meeting for the same shall be required. Till such time the connectivity of 50 MW to M/s Sunsire is proposed to be granted in sharing with M/s Proteus(217 MW). However same shall be finalized after above meeting only. It was also informed that M/s Sunsire Solarpark RJ One Private has requested Start Date of Connectivity under GNA from 31.03.2026. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 31.03.2026 (interim). M/s Sunsire noted the same. Transmission system for connectivity of Sunsire Solarpark Rj One Private under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u> (i). Through sharing of dedicated transmission system of M/s Proteus Energy Pvt. Limited– Bikaner-III PS 220 kV S/c line (Suitable to carry minimum 317 MW at nominal voltage) (220 kV bay at Bikaner-III PS shall be common for Proteus(217 MW)+M/s Avaada(50 MW) & M/s Sunsire(50 MW))											

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
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C. Common Transmission system for Connectivity under GNA

- 1) Augmentation of 1x500 MVA (4th) 400/220 kV ICT at Bikaner-III Pooling Station
- 2) Augmentation of 2x1500 MVA (5th & 6th) 765/400 kV ICTs at Bikaner-III Pooling Station
- 3) Augmentation of 1x1500 MVA (4th) 765/400kV ICT at Bikaner (PG)
- 4) LILO of both ckts of 400 kV Bikaner(PG)-Bikaner-II D/c line (Quad) at Bikaner-III PS
- 5) Bikaner-II PS – Bikaner-III PS 400 kV D/c line (Quad)
- 6) Bikaner-III - Neemrana-II 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at each end
- 7) Augmentation of 2x1500 MVA (3rd & 4th),765/400 kV ICTs at Neemrana-II S/s
- 8) Neemrana-II -Kotputli 400 kV D/c line (Quad)
- 9) LILO of both ckts of 400 kV Gurgaon (PG) - Sohna Road (GPTL) D/c line (Quad) at Neemrana-II S/s
- 10) Bikaner-III - Neemrana-II 765 kV D/c line (2nd) along with 330 MVAR switchable line reactor for each circuit at each end
- 11) Neemrana-II- Bareilly (PG) 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at each end
- 12) Augmentation by 400/220 kV, 1x500 MVA ICT (3rd) at Kotputli (PG)
- 13) 400 kV Bareilly(765/400kV) – Bareilly(PG) D/c line (Quad) (2nd)
- 14) Augmentation with 1x1500 MVA, 765/400 kV ICT at Bareilly (765/400kV) S/s (3rd)

D. Additional Transmission system for Inter-regional power transfer:

1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors
2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end
- 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line
- OR
- 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor
- 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line
- 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar

Start Date of Connectivity under GNA: 31.03.2026(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
3.	2200000210	Solarcraft Power India 17 Private Limited	Bikaner distt., Rajasthan	05.09.2023	Generator (Solar)	Land Route	300	30.12.2025	Bikaner-III PS	Bikaner-IV PS	Conn-BG1: Rs. 50 lakh Conn-BG2: Nil(bay under applicant scope) Conn-BG3: Rs.2 lakh/ MW

It was informed that M/s Solarcraft Power India 17 Pvt. Ltd. has applied for connectivity for 300 MW at Bikaner-III PS. As informed earlier, there is no margin available for grant of connectivity at Bikaner-III PS. Accordingly, it was proposed to grant connectivity at Bikaner-IV PS at 220 kV level through 1 no. of 220 kV line bay.

M/s Solarcraft confirmed that the 220kV bay at Bikaner-IV PS end shall be under applicant scope and the tower configuration shall be S/c on D/c tower. M/s Solarcraft was asked to submit necessary undertaking for tower configuration of S/c on D/c. M/s Solarcraft noted and agreed for the same.

It was also informed that M/s Solarcraft Power India 17 Pvt. Ltd. has requested Start Date of Connectivity under GNA from 30/12/2025. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 30.06.2026(interim). M/s Solarcraft noted the same. Transmission system for connectivity of M/s Solarcraft Power India 17 Pvt. Ltd under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

- (i). Solarcraft Power India 17 Private Limited – Bikaner-IV PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage)

C. Common Transmission system for Connectivity under GNA

- 1) Establishment of 2x1500 MVA, 765/400 kV & 2x500MVA (1st & 2nd) 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor at a suitable location near Bikaner
- 2) LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line (2nd) at Bikaner-IV PS along with 330 MVAR switchable line reactor at Bikaner-IV PS end of each ckt of 765 kV Bikaner-IV -Neemrana-II line (formed after LILO)
- 3) Establishment of 765/400 kV, 2x1500 MVA S/s at suitable location near Churu along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor
- 4) Bikaner-IV PS – Churu 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at Bikaner-IV PS end

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
5) LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s 6) Churu – Fatehabad (PG) 400 kV D/c line (Quad Moose) along with 63 MVAR switchable line reactor for each circuit at Churu S/s end 7) Establishment of 765/400 kV, 3x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVAR (765kV) Bus Reactor & 2x125 MVAR (420kV) Bus Reactor 8) Siwani – Jind (PG) 400 kV D/c line (Quad Moose) 9) Siwani – Patran (Indi Grid) 400 kV D/c line (Quad Moose) along with 80 MVAR switchable line reactor for each circuit at Siwani S/s end 10) STATCOM (2x+300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR) along with 2 nos. 400kV line bays at Bikaner-IV PS D. Additional Transmission system for Inter-regional power transfer: 1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors 2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 30.06.2026(Interim). Final date shall be confirmed upon award of the system											
4.	2200000244	NTPC Renewable Energy Limited	Bikaner distt., Rajasthan	07.09.2023	Generator (Solar)	LOA or PPA	150	28.06.2025	Bikaner-III PS	Bikaner-IV PS	Conn-BG1: Rs. 50 lakh Conn-BG2: NIL (In sharing) Conn-BG3: Rs.2 lakh/ MW
As discussed above for the application of 500 MW by NTPC(App No.2200000179), it was proposed to grant connectivity for cumulative quantum of 650MW (500MW+150MW) to NTPC Limited at Bikaner-IV PS at 400 kV level through 1 no. of 400 kV line bay. NTPC agreed for the same. It was also informed that Transmission scheme for Bikaner-IV PS is being taken up for discussion and approval in the present meeting. It was also informed that M/s NTPC Renewable Energy Limited has requested Start Date of Connectivity under GNA from 28/06/2025. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 30.06.2026 (interim). M/s NTPC noted the same. Transmission system for connectivity of NTPC Renewable Energy Limited under GNA is as below:											

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
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Details of Transmission system for Connectivity under GNA:**A. Associated Transmission System (ATS): NIL****B. Transmission System under applicant scope**

- (i). Common pooling station for NTPC Renewable Energy Limited Solar Power Projects(App No.2200000179(500 MW) & 2200000244(150 MW) – Bikaner-IV PS 400 kV S/c line on D/c tower (Suitable to carry minimum 900 MW at nominal voltage)

C. Common Transmission system for Connectivity under GNA

- 1) Establishment of 2x1500 MVA, 765/400 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor at a suitable location near Bikaner
- 2) LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line (2nd) at Bikaner-IV PS along with 330 MVAR switchable line reactor at Bikaner-IV PS end of each ckt of 765 kV Bikaner-IV -Neemrana-II line (formed after LILO)
- 3) Establishment of 765/400 kV, 2x1500 MVA S/s at suitable location near Churu along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor
- 4) Bikaner-IV PS – Churu 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at Bikaner-IV PS end
- 5) LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s
- 6) Churu – Fatehabad (PG) 400 kV D/c line (Quad Moose) along with 63 MVar switchable line reactor for each circuit at Churu S/s end
- 7) Establishment of 765/400 kV, 3x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor
- 8) Siwani – Jind (PG) 400 kV D/c line (Quad Moose)
- 9) Siwani – Patran (Indi Grid) 400 kV D/c line (Quad Moose) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end
- 10) STATCOM (2x+300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV line bays at Bikaner-IV PS

D. Additional Transmission system for Inter-regional power transfer:

1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors
2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVar 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVar switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 30.06.2026(Interim). Final date shall be confirmed upon award of the system											
5.	2200000232	Green Infra Clean Energy Generation Limited	Jaisalmer distt., Rajasthan	08.09.2023	Generator (Solar)	Land Route	300	01.06.2025	Bhadla-III PS	Bhadla-III PS	Conn-BG1: Rs. 50 lakh Conn-BG2: Rs. 3 Cr. Conn-BG3: Rs.2 lakh/ MW
It was informed that M/s Green Infra Clean Energy Limited has applied for connectivity for 300 MW at Bhadla-III PS. Accordingly, it was proposed to grant connectivity at Bhadla-III PS at 220 kV level through 1 no. of 220 kV line bay. M/s Green Infra informed that 220 kV bay at Bhadla-III PS end shall be under the scope of ISTS and the tower configuration shall be S/c on D/c tower. CTUIL asked M/s Green Infra to submit necessary undertaking for tower configuration of S/c on D/c. M/s Green Infra agreed for the same. It was also informed that M/s Green infra has requested Start Date of Connectivity under GNA from 01/06/2025. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 30.06.2028(interim). M/s Green Infra noted the same. Transmission system for connectivity of M/s Green Infra Clean Energy under GNA Regulations 2022 is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope -</u> (i). Green Infra Clean Energy Generation Ltd Solar Power Project – Bhadla-III PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage)											

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
C. Common Transmission system for Connectivity under GNA : - Augmentation of 1x500 MVA (8th) 400/220kV ICT at Bhadla-III PS - Augmentation of 2x1500 MVA (3rd & 4th), 765/400kV ICT at Bhadla-III PS 765 kV Bhadla-III – Bikaner-III D/c line along with 240 MVar switchable line reactor for each circuit at Bhadla-III end - Establishment of 6000 MW, $\pm$ 800 kV Bhadla (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near Bhadla-III substation - Establishment of 6000 MW, $\pm$800 kV Fatehpur (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Fatehpur (UP) - Bhadla-III – Bhadla (HVDC) 400 kV D/c (1st & 2nd) quad moose line along with the line bays at both substations - Bhadla-III – Sikar-II 765 kV D/c line along with 330 MVar Switchable line reactor for each circuit at each end of Bhadla-III – Sikar-II 765 kV D/c line - Fatehgarh-III – Bhadla-III 400kV D/c line (Quad) along with 63 MVar Switchable line reactor for each circuit at both ends of Fatehgarh-III- Bhadla-III 400kV D/c line - Augmentation of 5x1500 MVA (1st to 5th), 765/400 kV ICTs at Fatehpur (HVDC) along with 2x330 MVar (765kV) bus reactor - LILO of both ckts of 765 kV Varanasi – Kanpur (GIS) D/c line at Fatehpur - Augmentation by 1x1500 MVA ICT at 765/400kV Kanpur (GIS) substation											
Start Date of Connectivity under GNA: 30.06.2028(Interim) Final date shall be confirmed upon award of the system.											
6.	2200000221	First Energy Private Limited	Bikaner distt., Rajasthan	11.09.2023	Captive Generating Plant (Solar)	Land BG Route	100	31.05.2025	Bikaner-II PS	Bikaner-IV PS	Conn-BG1: Rs. 50 lakh Conn-BG2: Nil (under applicant scope) Conn-BG3: Rs.2 lakh/ MW
It was informed that M/s First Energy Private Limited has applied for connectivity for 100 MW at Bikaner-II PS. However, as informed in the 23rd CMETS NR meeting, 675 MW of margin available at Bikaner-II PS cannot be granted to any entity till further order by CERC. Further, any vacancy created at Bikaner-II PS shall also be offered to applicants at Bikaner-III PS as per application priority. As even at Bikaner-III PS, connectivity capacity is already exhausted, it was proposed to grant connectivity at Bikaner-IV PS. It was also informed that other connectivity grantees at Bikaner-IV PS are of 300-400MW and therefore no capacity margin is available in general on those bays. Therefore, even though M/s First Energy Private Limited has applied for only 100MW, in view of no margin available at other bays, it was proposed to grant connectivity at Bikaner-IV PS at 220 kV level through 1 no. of 220 kV line bay. It was also informed that balance margin (200MW) available in 220 kV bay shall be granted to other developers on sharing basis with M/s First Energy. M/ First Energy agreed for the same. M/s First Energy informed that tower configuration and bay scope shall be confirmed within 2-3 days. Subsequently, M/s First Energy vide mail dated 06/11/2023 confirmed their acceptance of connectivity at Bikaner-IV PS and informed that the 220 kV bay at Bikaner-IV PS shall be under their scope along with S/c line configuration.											

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
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It was also informed that M/s First Energy Private Limited has requested Start Date of Connectivity under GNA from 31/05/2025. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 30.06.2026(interim). M/s First Energy noted the same.

Transmission system for connectivity of M/s First Energy under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

- (i). First Energy Private Limited – Bikaner-IV PS 220 kV S/c line (Suitable to carry minimum 300 MW at nominal voltage)

C. Common Transmission system for Connectivity under GNA

- 1) Establishment of 2x1500 MVA, 765/400 kV & 1x500MVA (2nd) 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor at a suitable location near Bikaner
- 2) LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line (2nd) at Bikaner-IV PS along with 330 MVAR switchable line reactor at Bikaner-IV PS end of each ckt of 765 kV Bikaner-IV -Neemrana-II line (formed after LILO)
- 3) Establishment of 765/400 kV, 2x1500 MVA S/s at suitable location near Churu along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor
- 4) Bikaner-IV PS – Churu 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at Bikaner-IV PS end
- 5) LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s
- 6) Churu – Fatehabad (PG) 400 kV D/c line (Quad Moose) along with 63 MVar switchable line reactor for each circuit at Churu S/s end
- 7) Establishment of 765/400 kV, 3x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor
- 8) Siwani – Jind (PG) 400 kV D/c line (Quad Moose)
- 9) Siwani – Patran (Indi Grid) 400 kV D/c line (Quad Moose) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end
- 10) STATCOM (2x300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV line bays at Bikaner-IV PS

D. Additional Transmission system for Inter-regional power transfer:

1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line OR 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAR 765 kV bus reactor 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar Start Date of Connectivity under GNA: 30.06.2026(Interim). Final date shall be confirmed upon award of the system											
7.	2200000285	Sunsure Solarpark Fourteen Pvt Ltd	Bikaner distt., Rajasthan	30.09.2023	Generator (Solar)	Land BG Route	300	31.03.2027	Bikaner-III PS	Bikaner-IV PS	Conn-BG1: Rs. 50 lakh Conn-BG2: Rs. 3 Cr. Conn-BG3: Rs.2 lakh/ MW
It was informed that M/s Sunsare Solarpark Fourteen Private Limited has applied for connectivity for 300 MW at Bikaner-III PS. As informed earlier, there is no margin available for grant of connectivity at Bikaner-III PS. Accordingly, , it was proposed to grant connectivity for at Bikaner-IV PS at 220 kV level through 1 no. of 220 kV line bay. It was informed that transmission scheme for Bikaner-IV PS is under planning and is being taken up in present meeting for approval. M/s Sunsare informed that 220 kV bay at Bikaner-IV PS end shall be under the scope of ISTS and the tower configuration shall be S/c on D/c tower. CTUIL asked M/s Sunsare to submit necessary undertaking for tower configuration of S/c on D/c. M/s Sunsare agreed for the same. It was also informed that M/s Sunsare Solarpark Fourteen Private Limited has requested Start Date of Connectivity under GNA from 31.03.2027. Considering the same and expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 31.03.2027 (interim) as requested. M/s Sunsare noted and agreed for the same. Transmission system for connectivity of Sunsare Solarpark Fourteen Private Limited under GNA is as below: <u>Details of Transmission system for Connectivity under GNA:</u> A. <u>Associated Transmission System (ATS):</u> NIL B. <u>Transmission System under applicant scope</u>											

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
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(i). Sunsore Solarpark Fourteen Private Limited– Bikaner-IV PS 220 kV S/c line on D/c towers (Suitable to carry minimum 300 MW at nominal voltage)

C. Common Transmission system for Connectivity under GNA

- 1) Establishment of 2x1500 MVA, 765/400 kV & 1x500MVA(2nd) 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor at a suitable location near Bikaner
- 2) LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line (2nd) at Bikaner-IV PS along with 330 MVAR switchable line reactor at Bikaner-IV PS end of each ckt of 765 kV Bikaner-IV -Neemrana-II line (formed after LILO)
- 3) Establishment of 765/400 kV, 2x1500 MVA S/s at suitable location near Churu along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor
- 4) Bikaner-IV PS – Churu 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at Bikaner-IV PS end
- 5) LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s
- 6) Churu – Fatehabad (PG) 400 kV D/c line (Quad Moose) along with 63 MVar switchable line reactor for each circuit at Churu S/s end
- 7) Establishment of 765/400 kV, 3x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor
- 8) Siwani – Jind (PG) 400 kV D/c line (Quad Moose)
- 9) Siwani – Patran (Indi Grid) 400 kV D/c line (Quad Moose) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end
- 10) STATCOM (2x+300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV line bays at Bikaner-IV PS

D. Additional Transmission system for Inter-regional power transfer:

1. Establishment of 765kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors
2. Beawar- Mandsaur PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end
- 3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line
- OR
- 3b) Establishment of 765 kV Kurawar S/s with 2x330 MVar 765 kV bus reactor
- 3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVar switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line
- 3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar

Start Date of Connectivity under GNA: 31.03.2027(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
8.	2200000286	Enren-I Energy Private Limited	Barmer distt., Rajasthan	30.09.2023	Generator (Solar)	Land BG Route	300	25.02.2026	Barmer-I PS	Barmer-I PS	Conn-BG1: Rs. 50 lakh Conn-BG2: Rs. 3 Cr. Conn-BG3: Rs.2 lakh/ MW

It was informed that M/s Enren-I Energy Private Limited has applied for connectivity for 300 MW at Barmer-I PS. Accordingly, it is proposed to grant connectivity at Barmer-I PS at 220 kV level through 1 no. of 220 kV line bay.

M/s Enren-I informed that 220 kV bay at Barmer-I PS end shall be under the scope of ISTS and the tower configuration shall be S/c on D/c tower. CTUIL asked M/s Enren-I to submit necessary undertaking for tower configuration of S/c on D/c. M/s Enren-I agreed for the same.

It was also informed that M/s Enren-I Energy Private Limited has requested Start Date of Connectivity under GNA from 25/02/2026. However, considering the expected commissioning date of transmission system, the start date of Connectivity under GNA shall be 28.02.2026(interim). M/s Sunsure noted the same.

Transmission system for connectivity of M/s Enren-I Energy Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

- (i). Enren-I Energy Private Limited Solar Power Project – Barmer-I PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage)

C. Common Transmission system for Connectivity under GNA

- 1) 2x1500 MVA (2nd & 3rd), 765/400 kV ICTs at Barmer-I Pooling Station
- 2) 1x500 MVA (3rd), 400/220 kV ICT at Barmer-I Pooling Station
- 3) Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad)
- 4) Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVAR (765 kV) & 2x125 MVAR (400 kV) Bus Reactor
- 5) Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end
- 6) Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVAR switchable line reactor for each circuit at Sirohi PS end (Quad)

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Proposed RE Pooling Station/ Substation for Grant of Connectivity	Conn BGs to be submitted as per GNA Regulations
7)										Establishment of, 765 kV Substation at suitable location near Rishabdeo (Distt. Udaipur) along with 2x240 MVAR (765 kV) Bus Reactor	
8)										Sirohi PS- Rishabdeo 765 kV D/c line along with 330 MVAR switchable line reactor for each circuit at Sirohi end	
9)										Rishabdeo - Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at Rishabdeo end	
10)										LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo S/s	
11)										Establishment of 765 kV Mandsaur PS along with 2x330 MVAR (765 kV) Bus Reactors	
12)										Mandsaur PS–Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line	
13)										Establishment of 765/400 kV (2x1500 MVA Kurawar S/s (with 1x500 MVA spare single phase transformer unit) with 2x330 MVAR 765 kV bus reactor and 1x125 MVAR 420 kV bus reactor (with 1x110 MVAR & 1x80 MVAR, 765 kV spare single phase reactor unit for line/bus reactor)	
14)										Mandsaur – Kurawar 765 kV D/c line along with 240 MVAR switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line	
15)										LILO of Indore – Bhopal 765 kV S/c line at Kurawar	
16)										Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line	
17)										LILO of one circuit of Indore – Itarsi 400 kV D/c line at Astha	
18)										Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line	
Start Date of Connectivity under GNA: 28.02.2026(Interim). Final date shall be confirmed upon award of the system											

It was informed that as per 8.2(c) of GNA Regulations, Conn-BG1, Conn-BG2 and Conn-BG3, as applicable, shall be furnished within 1 (one) month of intimation of in-principle grant of Connectivity, failing which the application for Connectivity shall be closed and application fee shall be forfeited. Accordingly, all applicants were requested to submit requisite BGs within the stipulated time as mentioned above. Subsequent treatment of Conn-BG1, Conn-BG2 and Conn-BG3 shall be in terms of Regulations 16.1 to 16.4 of GNA regulations.

It is also informed that applicants who have opted for S/c on D/c tower configuration need to give necessary undertaking to CTUIL regarding the same.

2A. Application for Grant of GNA/GNA_{RE} to entities other than STU under Regulation 20.1, 20.3 and 20.4 to entities under Regulation 17.1(ii), (iii), (v) and (vi):

SI No.	Application ID	Name of the Applicant	Submission Date	Region	Nature of applicant	GNA within Region (MW)	GNA outside Region (MW)	Total Quantum (MW) of GNA Required	Start date of GNA	End date of GNA
1	2200000269	Tata Steel Limited	26-09-2023	NR	Drawee entity connected to Intra State Transmission System	0	46.12	46.12	01-10-25	31-05-50

It was informed that M/s Tata Steel Limited applied for GNA_{RE} of 46.12MW (outside the region) from 01.10.25 to 31.05.50 for drawl of power from 400/220kV Dhanansu S/s (PSTCL) at 220kV level. M/s Tata Steel also submitted NoC from PSTCL vide letter dated 25.09.23 for 48MW (max ceiling allowed) drawl of power from 01.06.25 to 31.05.50.. PSTCL has also granted LTA in its intra state system to M/s Tata Steel EAF plant at Kadiana Khurd, Ludhiana for drawl of upto 48MW power for captive use under inter state transaction from its solar and wind plants located in Maharashtra, Tamil Nadu and Karnataka . As per above letter 220kV line from Dhanansu substation of PSTCL and Tata EAF Plant is under implementation.

Further, the NOC granted by PSTCL is conditional which is subject to compliance of terms and conditions imposed vide PSTCL Conditional approval letter bearing Memo no. 1677 dated 15.09.2023 issued to Tata Steel Ltd, which also include requirement of commissioning of 220kV line from Dhanansu-Tata Steel EAF plant etc.

It was stated that Punjab has been granted GNA of 5497 MW as per 18.1 of GNA regulation. From the studies, it is observed that, there are sufficient margins in the existing transmission system for drawl of additional 46.12 MW of power by Punjab through ISTS network. In view of that, it is proposed to grant 46.12 MW of GNA_{RE} (outside the region) to Tata Steel from 01.10.25 to 31.05.50 for drawl of power from ISTS network.

It is to inform that entities granted GNA_{RE} should satisfy conditions as per the CERC ((Sharing of Inter-State Transmission Charges and Losses) Regulations, 2020(including the amendments) to be eligible for waiver of transmission charges.Further, as per 20.4 of GNA Regulations if an entity granted GNA_{RE} intends to draw power from the sources other than the sources specified in clause (2) of the Regulation 13 of the Sharing Regulations, it may:

- a. get such power scheduled under GNA_{RE} with payment of transmission charges @ Transmission deviation rate in accordance with the Sharing Regulations; or
- b. convert full GNA_{RE} into GNA by making an application to the Nodal Agency.

CTU enquired from SLDC, Punjab that whether NoC is issued by Punjab SLDC or STU. Punjab SLDC confirmed that NoC is issued by STU as Dy. Chief Engineer (Open access) is a nodal officer under STU for open access for issuance of such NOC. CTU enquired about schedule of dedicated transmission line as NoC is conditional include requirement of commissioning of dedicated line. In reply, Punjab SLDC confirmed that tentative schedule of 220kV line is Oct'24. Regarding load requirement and line configuration, M/s Tata Steel stated that line is 220kV double circuit line and their total requirement is about 100MVA (peak requirement), out of which 46.12MW will be meet from inter state green power and balance from DISCOM.

M/s Tata Steel stated that they have applied under regulation 17.1 (ii) of GNA regulation, there will be no requirement of any type of BGs. CTU informed that in requirement of BG will be done as per regulation. No other observation received from Punjab SLDC/STU. Accordingly, it was agreed to grant GNA_{RE} of 46.12MW to M/s Tata Steel (outside the region) from 01.10.25 to 31.05.50 for drawl of power from Punjab STU i.e, 400/220kV Dhanansu S/s (PSTCL) at 220kV level.

4. Transition of Connectivity granted to M/s Hindustan Zinc Limited (HZL) (200 MW) under GNA Regulations

S. No	Name of Applicant	Connectivity Quantum (MW)	System for GNA	Start Date Requested	Start date of Connectivity/GNA	Bank Guarantee Requirement as per GNA Regulation
1	Hindustan Zinc Limited	200	Augmentation of 4th ICT at Kankroli (PG)	01-Jul-24	31-May-2025 (Interim)	As per 22.2 (b-i) Regulation, Conn BG1: Rs. 50 lakh Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Hindustan Zinc Limited (HZL) was granted connectivity of 200 MW through 220 kV D/c line at Kankroli(PG) under Connectivity Regulations, 2009. Subsequently, HZL had opted for transition of application under GNA Regulations and requested for grant of GNA_{RE}. The transition of HZL under GNA Regulations was discussed in 23rd CMETS NR meeting held on 29.08.2023 wherein the HZL was asked to submit the bifurcation of 200 MW quantum in within & outside region. It was informed that based on the bifurcation, the system will be finalized and transition of HZL Connectivity into GNA_{RE} shall be processed.

Subsequently, CERC vide order dated 22.09.2023 in Petition No. 11/SM/2023 clarified the requirement of bifurcation of connectivity as within & outside region as below:

“24. In light of difficulties brought to our notice by CTUIL, we clarify that a drawee DIC is eligible to request a schedule from anywhere in India up to its GNA quantum, where such injection point may be “within the region” or “from outside the region”. Once such entity has placed a scheduling request with RLDC and there is a constraint in the transmission system due to which a full schedule as requested by all drawee DICs in the region cannot be accommodated, RLDC shall allocate the transmission corridor as follows:

(a) In case of constraint in the transmission system “from outside the region”, the transmission corridor shall be allocated in proportion to the “outside the region” bifurcation of all such drawee DICs.

(b) In case of constraint in the transmission system “within the region”, the transmission corridor shall be allocated in proportion to the total GNA quantum for such drawee DICs (sum of “within the region” and “from outside the region” bifurcation)”

In view of the above order, HZL vide email dated 26.09.2023 submitted their bifurcation for 200 MW GNA_{RE} as below:

- Within region- 5 MW
- Outside region- 195 MW

In view of the above, it was proposed to grant 200 MW of GNA_{RE} to M/s HZL at Kankroli (PG) S/s with Start date of GNA_{RE} from 31.05.2025 (interim). Final Start date of GNA_{RE} shall be confirmed upon award of identified transmission system.

Details of Transmission System for Grant of GNA_{RE}:

A. Transmission system for GNA_{RE} (under scope of Applicant and to be constructed and maintained by licensee):

- 220 kV Hindustan Zinc Switchyard - Kankroli D/c line along with bays at both ends

B. Transmission system for GNA_{RE} (under ISTS):

- Augmentation of 1x500 MVA 400/220 kV ICT(4th) at Kankroli (PG)
- Shifting of 50MVAR Bus Reactor at future bay to accommodate above ICT.
- 400 kV bay for termination of Shifting of bus reactor.

Start date of GNA_{RE}: 31.05.2025 (interim) (Considering 18 months schedule for augmentation of ICT at Kankroli(PG))

As per 12.5 of GNA Regulations, the line to connect HZL to ISTS and necessary augmentation for providing connection to the ISTS, shall be constructed and maintained by a licensee at the cost of HZL.

It was also highlighted that, as discussed in the 23rd CMETS NR, ICT augmentation at Kankroli (PG) shall cater to load of HZL as well as future incremental drawl requirement of RVPN from existing as well as future (Nathdwara) system. Therefore, the augmentation of ICT shall be considered as common augmentation and shall be under ISTS.

The augmentation of 4th ICT at Kankroli (PG) was approved in the 24th CMETS NR meeting held on 20.09.2023 wherein it was decided that the implementation of ICT shall be taken up(with schedule of 18 months) after award of Nathdwara system by RVPN or submission of requisite Conn BGs by M/s HZL for transition of 200 MW whichever is earlier.

RVPN informed that for Nathdwara S/s LOA is issued on 3rd Oct'23 and expected schedule is Apr'25. However, for the 220 kV line, they are going for rebidding for which bid opening is on 30.11.2023. Accordingly, in view of the requirement for HZL, CTU shall be taking up implementation of ICT subject to submission of Conn BGs by HZL. Accordingly, HZL was requested to expedite the submission of requisite Conn BGs to take up implementation of ICT. M/s HZL agreed for the same.

5. Application for grant of additional GNA by UPPTCL(STU):

It was informed that Uttar Pradesh(UP) has been granted deemed GNA of 10165 MW as per Annexure-I of GNA regulations and as per 18.1(e) of GNA Regulations, UP SLDC was to segregate the GNA of 10165 MW to each intra state entity. Accordingly, UP SLDC segregated the 10165 MW of GNA as below:

S. No.	Name of Entity	GNA (in MW)	GNA Within Region	GNA Outside Region
1	NPCL	351		351
2	Railway	178		178
3	UPPCL	9636	6526	3110
Total		10165	6526	3639

Further, as per 19.1 of GNA Regulations:

“Within 3 months of coming into force of these regulations, STU may, on behalf of intra-State entities including distribution licensees, apply for additional GNA over and above the GNA deemed to have been granted under clauses (a) to (e) of Regulation 18.1 of these regulations (with entity-wise segregation) indicating GNA within the region and from outside the region.”

CERC vide order dated 01.10.2023 on petition no. 15/SM/2023 allowed STUs to apply for additional GNA for any quantum till 31.03.2024, which shall be considered for grant by CTUIL in terms of the GNA Regulations.

Subsequently, UPPTCL(STU) on behalf of Railway submitted application for additional GNA of 174 MW(outside the region) on 11.10.2023 with details (under 7(a) of application) as below:

S. No.	Additional GNA Requested	GNA Within Region	GNA Outside Region	Start Date of GNA Requested	Peak Demand
1	Year-1 :174 MW	-	174	10.10.2023	28284
2	Year-2 :174 MW	-	174	01.04.2024	30500
3	Year-3 :174 MW	-	174	01.04.2025	32500

As per the 7(b) of application, UPPTCL has requested for incremental GNA of 174 MW every year. However, under entity wise segregation of GNA mentioned in 7(b) of application, the same GNA quantum is mentioned for each year as below:

Entity	Year-1	Year-2	Year-3
NPCL	351	351	351
Railway	352	352	352
UPPCL	9636	9636	9636
Total	10339	10339	10339

Considering above information in application(7(a) & 7(b)), it was understood that UPPTCL need additional GNA of 174 MW to Railways (178+174=352 MW) on one time basis and not 174 MW of GNA on incremental basis every year. UPPTCL clarified that they need 174 MW of GNA on one time basis only for Railway. Accordingly, total GNA for Railway shall be 352 MW.

It was informed that, there are sufficient margins in the existing transmission system for grant of additional GNA of 174 MW under Regulation 22.1(a) to UP. Accordingly, it was proposed to grant additional GNA of 174 MW (outside the region) to UP.

Further, it was informed that UPPTCL has requested start date of GNA from 10.10.2023. However, as the date has already elapsed, it was proposed that the GNA shall be granted from 10.11.2023 considering the time required for approval and issuance of intimation. M/s UPPTCL agreed for the same. Accordingly, it was decided to grant additional GNA of 174 MW (outside the region) to UP from 10.11.2023 through existing system.

6. Transition of Budhil HEP(70 MW) under 37.6(1)

It was informed that M/s Greenko Budhil HEP(70 MW) has been granted deemed GNA of 61.6 MW as per 18.1 of GNA Regulations. The same was also communicated to M/s Greenko vide CTUIL letter dated 02.11.2022. For the balance 8.4 MW of connectivity, M/s Greenko was required to apply for additional GNA under 17.2 as per 37.6(1) of GNA Regulations. Subsequently, M/s Greenko Budhil applied for additional GNA of 8.4 MW on 29.09.2023.

CERC vide notification dated 03.08.2023 had informed that Scheduling and Despatch of electricity based on quantum of GNA, GNA_{RE}, T-GNA and T-GNA_{RE} w.e.f 01.10.2023. Accordingly, to avoid any bottling up of power, till the processing of their transition request, provisional GNA of 8.4 MW was granted to Greenko Budhil HEP by CTUIL upon submission of Conn-BG3 of Rs.2 lakh/MW as per Para No. 77 of CERC order dated 22/09/2023 in the Petition No. 11/SM/2023

Budhil HEP is an existing project connected to ISTS at Chamera-III through 220 kV D/c line. From the studies it is observed that there is sufficient margin available in existing system beyond Chamera-III for evacuation of 8.4 MW. In view of above, it was agreed to grant additional GNA of 8.4 MW M/s Greenko and Intimation for grant of Connectivity shall be issued with the following Details:

S. No	Name of the Applicant	Additional GNA applied for (MW)	GNA to be granted through	Start Date of GNA requested	Actual Start Date of GNA	Conn BG Requirement as per GNA Regulations
1	M/s Greenko Budhil Hydro Power Pvt. Ltd.	8.4	Existing System beyond Chamera Pooling Station	Immediately	01.10.2023*	Rs. 2 lakh/MW (Already submitted as per CERC order dated 22/09/2023 in the Petition No. 11/SM/2023)

* The start date of GNA shall be subject to signing of Connectivity Agreement under GNA Regulations, 2022.

7. Network Expansion Scheme

7.1 Transmission system for evacuation of power from Bikaner Complex as part of Rajasthan REZ Ph-IV (Part-3 :3.6GW (Total 6 GW))

It was informed that Renewable Energy Zones (REZs) were identified by MNRE/SECI with a total capacity of 181.5 GW for likely benefits by the year 2030 in eight states, which includes 75 GW REZ potential in Rajasthan comprising of 15 GW Wind and 60 GW Solar. In this regard a Committee on Transmission Planning for RE was constituted by MOP for planning of the requisite Inter State Transmission System required for the targeted RE capacity by 2030. In this regard, a Comprehensive transmission plan for evacuation of 75GW RE potential from Rajasthan is evolved. Details of schemes approved scheme as part of above is as under:

S.No	Transmission Scheme	RE Potential	Status
1	Rajasthan REZ Ph-IV (Part-1 :7.7GW) (Bikaner Complex)	14 GW (Solar 14GW, BESS:6GW)	Under Advance stage of Bidding
2	Rajasthan REZ Ph-IV (Part-2 :5.5GW) (Jaisalmer/Barmer Complex)	14GW (Wind : 7GW, Solar:7GW, BESS: 3 GW)	Under Bidding

Transmission scheme is evolved for about 7.7GW (Solar) in Bikaner complex (14 GW potential along with 6 GW BESS) in Rajasthan for RE potential identified at Bikaner complex as part of committee report. However, no application of BESS (linked with RE) against envisaged 6GW was received. Accordingly, RE potential of about 7.7GW (in place of 14GW) can be evacuated from planned system (Ph-IV scheme) from Bikaner complex (Bikaner-II(3.7 GW) & Bikaner-III(4 GW)).

At Bikaner-II PS & Bikaner-III PS connectivity of about 7.7GW utilizing above Ph-IV (Part-1) system for transfer of power is already granted and no further margin is available for additional connectivity due to technical limitation. Further, additional Connectivity of about 1.7 GW RE (Solar) is also received at Bikaner complex and more applications are expected due to land availability and being outside of GIB area for which new pooling station i.e. Bikaner-IV and onwards 765kV high capacity corridors will be required.

It was stated that Joint Study Meeting(s) were held in virtual mode on 18.10.23 and 23.10.23 with SECI, CEA, GRID-INDIA, RVPN, HVPN, PSTCL and other STUs of Northern region to finalize the Transmission system for evacuation of power from Bikaner Complex as part of Rajasthan REZ Ph-IV (Part-3). In the meeting, SECI/MNRE were also asked to confirm on RE potential of Bikaner complex.

In the first meeting, SECI stated that as part of committee report, RE potential of 30GW (out of 75GW) was planned in non GIB areas of Rajasthan viz. Bikaner, Jalore, Santhore, Sirohi, Ajmer etc. in phase-II & III scheme (except in Bikaner which was in Ph-I also), however Bikaner complex has much more RE potential and waste land availability out of above Non GIB zones. Further 6GW BESS at Bikaner-II/III PS could not be developed. SECI stated that at present 75GW is not yet fully harnessed and therefore adjustment of RE potential may be considered at present within Non-GIB area and if required the potential of 75 GW will be reviewed and increased in consultation and approval with MNRE later on.

CEA stated that as informed by CTU that no margin available for connectivity in Bikaner complex, potential for new substation at Bikaner-IV needs to be frozen for next phase of planning. CEA enquired SECI on progress of new bids for RE with BESS. SECI stated that at present there is no clear visibility for such projects (RE with BESS) before 2027 as award process will take time (1-2 years). CEA mentioned that 6 GW RE potential (solar) remains untapped due to non-materialisation of BESS capacity. CTU also informed that at present no RE application are received from any of complexes in non GIB areas viz. Santhore, Jalore Pali etc.

SECI also stated that 6GW potential at Bikaner complex can be considered for now as part of 75GW potential in Rajasthan with some portion from adjustment of potential from other Non-GIB complexes i.e. Santhore, Jalore, Pali etc. and balance from 6GW unharnessed RE potential (with BESS) at Bikaner complex. Accordingly, it was decided that cumulatively 6 GW RE potential may be considered for planning of transmission scheme from Bikaner-IV PS.

In the above first joint study meeting, CTU stated that considering requirement of 6 GW evacuation capacity from Bikaner-IV PS, 3GW RE capacity to be evacuated through EHVAC system as part of present proposal and balance 3GW along with Bhadla-IV potential (4GW) through HVDC system, which is under planning. In view of that system studies were carried and proposed Transmission scheme is as under

Proposed Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-3)

Bikaner-IV : 3GW (Solar)

- Establishment of 3x1500 MVA, 765/400 kV & 3x500 MVA 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor at a suitable location near Bikaner *
- 220kV line bays (4 nos.) for RE connectivity at Bikaner-IV PS
- STATCOM (2x+300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV line bays at Bikaner-IV PS
- Establishment of 765/400 kV, 3x1500 MVA Hissar-II S/s along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor at a suitable location near Hissar*
- Bikaner-IV – Hissar-II 765 kV D/c line (~320 km) along with 330 MVar switchable line reactor for each circuit at each end
- Bikaner-IV – Sikar-II 400 kV D/c line (Quad) (~270 km) along with 63 MVar switchable line reactor for each circuit at each end
- Hissar-II- Jind(PG) 400 kV D/c line (Quad) (~50 km)
- Hissar-II- Patran 400 kV D/c line (Quad) (~110 km)

Study files for solar maximized scenario was shared on 13.10.23 with all constituents. Grid-India vide mail 18.10.23 sent their observations on proposal and studies. Same was also deliberated in meeting. Reply of major observation is under

- 1. Demands of Punjab, Haryana & Rajasthan seem to be on the lower side looking at demand observed in the current time-period.**

CTU stated that in winter scenario, demand of Punjab and Haryana is lesser, and they will review the demand and change it in revised files for 2026 time frame, however Rajasthan demand in winter and summer scenario is as per 20th EPS demand (load+losses).

2. N-1 non compliance of ICTs in Rajasthan

CTU stated that most of the 400/220kV ICTs which are N-1 non compliant are in intra state (STUs), some of which are already highlighted in OCC/NRPC meetings. Regarding ICTs at Kota (PG), high loading is observed due to no KTPS generation considered in study files. RVPN conformed that KTPS units will be operational by 2027-28 and decision on KTPS units to take out of the system will be taken by management. In view of that in revised files KTPS unit will kept on. Loading of Bassi ICT is marginally higher in N-1 (320MW). Same will be reviewed and taken up separately in consultation with Grid-India & CEA based on real time loading.

3. High fault current at 220kV buses

Regarding high fault level of some of RE injection stations (at 220kV level), CTUIL stated that fault level of 220 kV level at old RE generating pooling stations i.e. Bhadla(PG), Bhadla-II, Fatehgarh-II is marginally higher (designed for 40kA). It was informed that during fault level calculation fault level contribution from RE generation of designated 220kV bus is not considered. Same philosophy is also considered for SCR calculation. Grid-India stated that in case of distance fault, there is a possibility that RE generation will also contribute in fault. CTU stated that matter will be separately discussed with Grid-India as new proposed scheme is not influence the fault levels on these old RE pooling stations.

4. Angular difference between 765kV Bikaner-IV & Hissar

Grid-India stated that under N-2 contingency of 765kV Bikaner-IV – Hissar line (both ckts out), there are severe file convergence issues. CTUIL stated that in the event of N-2 contingency, there will be no path available for evacuation of 3GW power from Bikaner-IV PS. CTUIL stated that they have carried out various cases to provide anchoring to Bikaner-IV from nearby ISTS stations i.e. Bikaner(PG), Bikaner-II, Bikaner-III. In all scenarios, power will flow towards Bikaner-IV and evacuation system requirement will increase substantially in base case.

Based on deliberations, various other options were also carried out in first meeting i.e. LILO of Bikaner-III – Neemrana-II D/c at Bikaner-IV, Bikaner-IV-Sikar-II 765kV D/c etc. CTU stated that with LILO of Bikaner-III – Neemrana-II D/c at Bikaner-IV case is converged in 'N-2' contingency but angular separation will be more than 30 degree.

In the meeting, CTUIL stated that critical loading is observed in 220kV Patran – Patran (PSTCL) D/c line in studies. PSTCL stated that in future Peddy scenarios, line loading will be higher and N-1 non compliant. CTU stated that even without injection at Patran from above scheme, 220kV Patran – Patran (PSTCL) D/c line remains N-1 non compliant and PSTCL may take suitable measures to relieve the loading. CTU stated that

ownership of this intra state line is with STU (PSTCL), accordingly, suitable strengthening needs to be planned and implemented by PSTCL as part of intra state scheme. PSTCL stated that they will revert on the same.

Based on Grid-India comments, CTU stated that N-1-1 or N-2 is a rare contingency and deliberations are required on N-1-1/N-2 compliance in planning studies in reference to manual on transmission planning criteria 2023. In view of above in the first meeting it was concluded that that other options may be explored by CTU in consultation with CEA and Grid-India and will be discussed in next Joint study meeting.

In the 2nd joint study meeting held on 23.10.23, CTU stated that they have explored various other alternatives and in new proposal, an intermediate substation in 765/400kV Churu is created with its connectivity to LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s and Fatehabad (PG) through 400 kV D/c line. To provide anchoring at Bikaner-IV, LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line (2nd) at Bikaner-IV PS is also considered. With above revised proposal, power flow is in order and angular separation incl. in N-2 contingency and voltages are within limit. Revised Study files for solar maximized scenario was shared with all constituents on 20.10.23.

Grid-India stated that the proposed system is optimal & balanced and loading is equally distributed among various feeders. The proposed system is stable and comply the N-1 & N-1-1 requirements. CEA stated that as per transmission planning criteria 2023, under N-1-1, some of the equipment may be loaded up to their emergency limits. To bring the system parameters back within their normal limits, load shedding/re-scheduling of generation may have to be done, either manually or through automatic system protection schemes (SPS).

In view of that in planning stage, N-1-1 or N-2 criteria may not be considered except in critical lines (Inter regional corridors) as it will increase the transmission system requirement. Therefore, in revised proposal a direct interconnection between Bikaner-IV and Siwani may be considered and 400kV interconnection towards Fatehabad may be planned from Siwani in place of via Churu.

CTU stated that in above alternative with direct interconnection to Siwani, file is converged, and power flow is in order, however angular separation between Siwani and Bikaner-IV is more than 50 degree in N-2 contingency in case of direct interconnection of Bikaner-IV and Siwani. The angular separation would be more than 30 degree even with significant less RE capacity (<1GW) at Bikaner-IV PS.

Grid-India stated that it is not recommended that SPS implementation is considered at planning stage. SPS requirement will generally come when study assumption considered in studies during planning may deviate at later stage i.e. load change or delay in interlinked transmission system which influence the load flow.

Further, in case of direct interconnection of Bikaner-IV to Siwani i.e. not considering 765/400kV Churu in between and onward transmission system, the Transmission system is kind of radial system connected with RE generation pocket and poses stability issues in various operational scenarios in future. Grid-India also emphasised that some margin should be kept in planning studies for operational scenarios. Further, the

angular difference (>30 degrees) in N-1-1 /N-2 may cause problems in synchronization of lines after corridor outage. The proposed system is stable and well interconnected with Grid provides reliable power evacuation under various operational scenarios

CEA stated that N-1-1 or N-2 is a rare contingency and deliberations are required on N-1-1/N-2 compliance in planning studies as it may incur additional investment for strengthening of transmission system.

CTU stated that in above case the transmission scheme i.e. establishment of Churu substation along with LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s and Churu – Fatehabad (PG) 400 kV D/c line will improve the system resiliency and address the Grid India concern for synchronization of lines after corridor outage in N-1-1 /N-2) contingency. Considering 3.6GW generation at Bikaner-IV PS, loading and angular separation in contingency is as under:

Transmission Line	N-1 (Loading)	N-1-1 /N-2 (Degree)
765 kV Bikaner-IV-Churu D/c line	14.1 (3098 MW)	29.2
765 kV Churu- Siwani D/c line	7.7 (2743 MW)	28.6

CTU further stated that in next phase, EHVAC/HVDC system (5-6GW) for evaluation of RE power from Bhadla-IV potential & balance potential of Bikaner-IV (~2.4GW) towards UP/outside NR region is under planning and will be taken up in subsequent meetings. CTU requested CEA to convene a joint meeting for compliance of N-1-1/N-2 in planning studies in reference of planning criteria 2023 by next week. CEA agreed for same. PSTCL stated that they will take suitable measures to relieve loading of 220kV Patran – Patran (PSTCL) D/c line in matching timeframe of above agreed ISTS scheme. **Minutes of joint study meeting is enclosed in Anenxure-2.**

Further the issue was deliberated in the CEA meeting with CTU and Grid-India held on 30.10.23 in which philosophy and applicability for consideration of N-1-1/N-2 in planning studies in reference to Manual on transmission planning criterion 2023 was discussed. In the meeting it was concluded that issue will be further deliberated with Member (PS) and accordingly scheme will be finalized in CMETS-NR meeting.

Subsequently, in the CMETS-NR meeting, CEA opined that Bikaner-IV S/s is planned for 6GW evacuation and at present direct interconnection between Bikaner-IV and Siwani is to be planned in first phase (3GW) and 765/400kV Churu S/s along with associated transmission scheme may be taken up in later stage while planning of balance 3 GW transmission scheme. Grid-India further emphasised that in case Churu S/s to be taken up at later stage of planning, 765kV Bikaner-IV-Siwani D/c line may not able to synchronize after corridor outage (765kV Bikaner-IV-Siwani D/c line) due to higher angular difference (>30 degrees) in N-1-1/N-2 in peak solar hours. Grid-India stated that due to potential delays in the restoration of the 765 kV Bikaner-IV to Siwani D/c line, synchronization of lines might not be feasible during peak solar hours. Instead, this process might need to be deferred to non-solar hours, which would entail running the system in a depleted condition for several hours.

In view of Grid-India views, CTU stated that Bikaner-IV S/s is planned for evacuation of 6GW generation, out of which 3.6GW power will be evacuated through proposed system (via Churu) in first phase and in subsequent phase (Ph2) transmission system will be planned for evacuation of 2.4GW. CTU stated that one no. of high capacity corridor is provided under present scheme towards Punjab/Haryana to meet the demand requirement. In next phase of planning (for balance 2.4GW potential at Bikaner-IV PS) transmission scheme to be planned towards UP/other regions. CTU stated that they have already received about 1.7GW of connectivity applications in last 2 months at Bikaner-IV and with more applications, it is expected that new system (ph-2) need to be planned within 2-3 months. Considering above, it is envisaged that 765/400kV Churu S/s along with associated transmission scheme may also be utilized in next phase to facilitate evacuation of power along with other new corridors.

As time difference between implementation of these two schemes will be lesser (2-3 months), it is recommended that 765/400kV Churu S/s along with associated transmission scheme shall be taken up in Ph-I as part of present proposed scheme as it will improve resiliency and angular stability under N-1-1/N-2 contingency. CEA also agreed for taking up the Bikaner-IV PS scheme via Churu & associated system in first phase.

Subsequently, line lengths were reviewed in reference to utilisation of Gati Shakti portal for transmission planning and accordingly reactive compensation was also modified, as per the requirement. **Result of system studies for proposed scheme is enclosed in Exhibit-1.**

Considering grant of connectivity under GNA to new RE generators in Bikaner RE potential complex as well as for evacuation of power beyond Bikaner-IV PS, following transmission scheme was agreed in CMETS-NR meeting for evacuation of power from Rajasthan REZ Ph-IV (Part-3) [Bikaner IV :3.6GW]

Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-3 :3.6GW)

Bikaner-IV: 3.6GW (Solar)

- Establishment of 4x1500 MVA, 765/400 kV & 4x500 MVA, 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner *

Future provisions at Bikaner-IV PS: Space for

- 765/400kV ICT along with bays- 2 no.
- 765 kV line bays along with switchable line reactors – 10 nos.
- 765kV Bus Reactor along with bay: 1 no.
- 400 kV line bays along with switchable line reactor –4 nos.
- 400 kV line bays–4 nos.

- 400/220kV ICT along with bays -6 nos.
- 400 kV Bus Reactor along with bay: 1 no.
- 400kV Sectionalization bay: 2 sets
- 220 kV line bays for connectivity of RE Applications -11 nos.
- 220kV Sectionalization bay: 3 sets
- 220 kV BC (2 nos.) and 220 kV TBC (2 nos.)

***along with provision of 80MVAR and 110MVAR spare reactor (Single phase)**

- 220kV line bays (5 nos.) for RE connectivity at Bikaner-IV PS
- 400kV line bays (1 no.) for RE connectivity at Bikaner-IV PS
- 220kV Sectionalization bay (1 set) along with BC (2 nos.) and 220 kV TBC (2 nos.) at Bikaner-IV PS
- 400kV Sectionalization bay (1 set) at Bikaner-IV PS
- STATCOM (2x+300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR) along with 2 nos. 400kV line bays at Bikaner-IV PS
- LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line (2nd) at Bikaner-IV PS (~20 km) along with 330 MVAR switchable line reactor at Bikaner-IV PS end of 765 kV Bikaner-IV -Neemrana-II line (formed after LILO)
- Establishment of 765/400 kV, 2x1500 MVA S/s at suitable location near Churu along with 2x240 MVAR (765kV) Bus Reactor & 2x125 MVAR (420kV) Bus Reactor

Future provisions at Churu S/s: Space for

- 765/400kV ICTs along with bays- 4
- 765 kV line bays along with switchable line reactors – 12
- 765kV Bus Reactor along with bay: 1 nos.
- 400 kV line bays along with switchable line reactor –8
- 400 kV Bus Reactor along with bays: 1 no.
- 400kV Sectionalization bay: 2 sets
- 400/220kV ICT along with bays -4 nos.**
- 220 kV line bays for drawl -4 nos. **
- 220kV Sectionalization bay: 2 sets **

***along with provision of 80MVAR spare reactor (Single phase)**

****Drawl requirement at Churu to be confirmed by RVPN**

- Bikaner-IV PS – Churu 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at Bikaner-IV PS end (~180 km)
- LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s (~90 km)
- Churu – Fatehabad (PG) 400 kV D/c line (Quad) along with 63 MVAR switchable line reactor for each circuit at Churu S/s end (~150 km)

- Establishment of 765/400 kV, 3x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor*

Future provisions at Siwani S/s: Space for

- 765/400kV ICT along with bays- 3
- 765 kV line bays along with switchable line reactors – 6
- 765kV Bus Reactor along with bay: 1 nos.
- 400 kV line bays along with switchable line reactor –10
- 400 kV Bus Reactor along with bays: 1 no.
- 400kV Sectionalization bay: 2 sets
- 400/220kV ICT along with bays -4 nos. **
- 220 kV line bays for drawl -4 nos. **
- 220kV Sectionalization bay: 2 sets **

***along with provision of 80MVar spare reactor (Single phase)**

****Drawl requirement at Siwani to be confirmed by HVPN**

- Siwani – Churu 765 kV D/c line (~95 km)
- Siwani – Jind (PG) 400 kV D/c line (Quad Moose) (~110 km)
- Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) (~160 km) (400kV GIS duct :700m) along with 80 MVar switchable line reactor for each circuit at Siwani S/s end

Tentative commissioning schedule : 24 months from allocation

Annexure-I

List of Participants of 25th Consultation meeting for Evolving Transmission Schemes in NR held on 31.10.223

CEA

Smt. Manjari Chaturvedi	Director
Shri Nitin Deswal	Deputy Director

NRPC

Shri Rajat Dixit	Assistant Executive Engineer
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SECI

Shri R.K.Agarwal	Consultant
Shri Himanshu Gulati	Senior Engineer

Grid India

Shri Gaurav Singh	Manager
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CTU

Shri Kashish Bhambhani	GM (CTU)
Shri Sandeep Kumawat	DGM (CTU)
Smt. Ankita Singh	Ch. Manager (CTU)
Shri R Narendra Sathvik	Manager (CTU)
Shri Madhusudan Meena	Engineer (CTU)

RVPN

Shri Sanjay Mathur	
--------------------	--

PSTCL/SLDC

Shri Nitin (PSTCL)

Smt. Silky Rani (Punjab SLDC)

UPPTCL

Shri Pankaj Saxena

Connectivity/GNA Applicants

Shri Yogesh Kumar Sanklecha	Acme Heergarh Powertech Private Limited & Acme Urja Two Private Limited
Shri Ajay Kumar Pradhan	Acme Heergarh Powertech Private Limited & Acme Urja Two Private Limited
Shri Lokesh Yadav	Acme Cleantech Solution Pvt. Ltd
Shri Shubham Roy	Deshraj Solar Energy Private Limited
Shri Shobhit Kumar	Deshraj Solar Energy Private Limited
Shri Vivek Kodesia	Eden Renewables Cadet Private Limited
Shri Rohit Singh	Eden Renewables Cadet Private Limited
Shri Vishal Saxena	Enren-I Energy Private Limited / Engie India
Shri Vineet Pandey	Enren-I Energy Private Limited
Smt. Vasundhara Sen	First Energy Private Limited
Shri Vishnu Khandelwal	Hindustan Zinc Limited
Shri Arzaan Dordi	Serentica Renewables India Private Limited
Shri Vivek Hooda	Green Infra Clean Energy Generation Limited

Shri Anish Pasrija	Greenko Budhil Hydro Power Pvt. Ltd
Shri Animesh Manna	NTPC Renewable Energy Limited
Shri Kumar Vipul	Radiant Star Solar Park Private Limited
Shri Mohit Jain	ReNew Solar Power Private Limited
Shri Ayush Jain	Solarcraft Power India 4 Private Limited, Solarcraft Power India 17 Private Limited
Shri Ashish Agrawal	Solarcraft Power India 4 Private Limited, Solarcraft Power India 17 Private Limited
Shri Sumit Kaushik	Sunsure Energy Private Limited
Shri Tushar Raj	Tata Steel Limited
Shri Kush Kumar	Tata Steel Limited
Shri Ajay Kumar	Tata Steel Limited



ANNEX-1

सेंट्रल ट्रांसमिशन यूटिलिटी ऑफ इंडिया लिमिटेड

(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)

(भारत सरकार का उद्यम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.

(A wholly owned subsidiary of Power Grid Corporation of India Limited)

(A Government of India Enterprise)

Ref: C/CTU/N/ REZ Ph-IV (Part-3)

Date: 10.11.23

As per distribution list

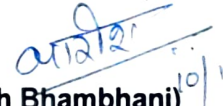
Sub: Minutes of Joint study meeting (s) held on 18.10.23 and 23.10.23 to finalize the Transmission system for evacuation of power from Bikaner Complex as part of Rajasthan REZ Ph-IV (Part-3)

Dear Sir,

Please find enclosed the minutes of Joint study meeting (s) held on 18.10.23 and 23.10.23 to finalize the Transmission system for evacuation of power from Bikaner Complex as part of Rajasthan REZ Ph-IV (Part-3) through virtual mode.

Thanking you,

Yours Faithfully,


(Kashish Bhambhani)
GM (CTU)

Encl : Minutes of Meeting

Distribution List:

Chief Engineer (PSP&A – I) Central Electricity Authority Sewa Bhawan, R.K.Puram, New Delhi-110 066	Member Secretary Northern Regional Power Committee 18A, Shaheed Jeet Singh Sansanwal Marg, Katwaria Sarai, New Delhi – 110 016
Director (Power System) Solar Energy Corporation of India Ltd. D-3, 1 st Floor, A wing, Religare Building, District Centre, Saket, New Delhi-110017	Director Ministry of New and Renewable Energy, Block 14, CGO Complex, Lodhi Road, New Delhi-110003
Director (SO) Grid Controller of India Limited (erstwhile Power System Operation Corporation Ltd.) 9 th Floor, IFCI Towers, 61, Nehru Place, New Delhi-110 016	Executive Director Northern Regional Load Despatch Centre 18-A, Qutab Institutional Area, Shaheed Jeet Singh Sansanwal Marg, Katwaria Sarai, New Delhi- 110 016
Director (P&C), HPPTCL, Headoffice, Himfed Bhawan, Panjari, Shimla-171005, Himachal Pradesh	Director(W&P) UP Power Transmission Company Ltd. Shakti Bhawan Extn, 3rd floor, 14, Ashok Marg, Lucknow-226 001
Director (Technical) Punjab State Transmission Corporation Ltd. Head Office, The Mall, Patiala 147001, Punjab	Director (Projects) Power Transmission Corporation of Uttarakhand Ltd. Vidyut Bhawan, Near ISBT Crossing, Saharanpur Road, Majra, Dehradun.
Development Commissioner (Power) Power Development Department Grid Substation Complex, Janipur, Jammu	Director (Technical) Rajasthan Rajya Vidyut Prasaran Nigam Ltd. Vidyut Bhawan, Jaipur, Rajasthan-302005.
Member (Power) Bhakra Beas Management Board Sector-19 B, Madhya Marg, Chandigarh - 160019	Superintending Engineer (Operation) Electricity Circle, 5 th Floor, UT Secretariat, Sector-9 D, Chandigarh - 161009
Director (Operations) Delhi Transco Ltd. Shakti Sadan, Kotla Road, New Delhi-110 002	Director (Technical) Haryana Vidyut Prasaran Nigam Ltd. Shakti Bhawan, Sector-6, Panchkula-134109, Haryana

Minutes of meeting of Joint study meeting (s) held on 18.10.23 and 23.10.23 to finalize the Transmission system for evacuation of power from Bikaner Complex as part of Rajasthan REZ Ph-IV (Part-3)

Joint Study Meeting(s) were held in virtual mode on 18.10.23 and 23.10.23 with SECI, CEA, GRID-INDIA, RVPN, HVPN, PSTCL and other STUs of Northern region to discuss the Transmission system for evacuation of power from Bikaner Complex as part of Rajasthan REZ Ph-IV (Part-3) scheme. In the meeting, SECI/MNRE were also asked to confirm on RE potential of Bikaner complex.

Gist of discussion held in First Joint study meeting on 18.10.23

In the meeting, It was informed that Renewable Energy Zones (REZs) were identified by MNRE/SECI with a total capacity of 181.5 GW for likely benefits by the year 2030 in eight states, which includes 75 GW REZ potential in Rajasthan comprising of 15 GW Wind and 60 GW Solar. In this regard a Committee on Transmission Planning for RE was constituted by MOP for planning of the requisite Inter State Transmission System required for the targeted RE capacity by 2030. In this regard, a Comprehensive transmission plan for evacuation of 75GW RE potential from Rajasthan is evolved. Details of schemes approved scheme as part of above is as under:

S.No	Transmission Scheme	RE Potential	Status
1	Rajasthan REZ Ph-IV (Part-1 :7.7GW) (Bikaner Complex)	14 GW (Solar 14GW, BESS:6GW)	Under Advance stage of Bidding
2	Rajasthan REZ Ph-IV (Part-2 :5.5GW) (Jaisalmer/Barmer Complex)	14GW (Wind : 7GW, Solar:7GW, BESS: 3 GW)	Under Bidding

CTU stated that Transmission scheme is evolved for about 7.7GW (Solar) in Bikaner complex (14 GW potential along with 6 GW BESS) in Rajasthan for RE potential identified at Bikaner complex as part of committee report. However, no application of BESS (linked with RE) against envisaged 6GW was received. Accordingly, RE potential of about 7.7GW (in place of 14GW) can be evacuated from planned system (Ph-IV scheme) from Bikaner complex (Bikaner-II(3.7 GW) & Bikaner-III(4 GW)).

At Bikaner-II PS & Bikaner-III PS connectivity of about 7.7GW utilizing above Ph-IV (Part-1) system for transfer of power is already granted and no further margin is available for additional connectivity due to technical limitation. Further, additional Connectivity of about 1.7 GW RE (Solar) is also received at Bikaner complex and more applications are expected due to land availability and being outside of GIB area for which new pooling station i.e. Bikaner-IV and onwards 765kV high capacity corridors will be required. In the meeting, SECI/MNRE were also asked to confirm on RE potential of Bikaner complex.

In the above meeting, SECI stated that as part of committee report, RE potential of 30GW (out of 75GW) was planned in non GIB areas of Rajasthan viz. Bikaner, Jalore, Santhore, Sirohi, Ajmer etc. in phase-II & III scheme (except in Bikaner which was in Ph-I also), however Bikaner complex has much more RE potential and waste land availability out of above Non GIB zones.

SECI stated that at present 75GW is not yet fully harnessed and therefore adjustment of RE potential may be considered at present within Non-GIB area and if required the potential of 75

GW will be reviewed and increased in consultation and approval with MNRE later on. Further 6GW BESS at Bikaner-II/III PS could not be developed.

CEA stated that as informed by CTU that no margin available for connectivity in Bikaner complex, potential for new substation at Bikaner-IV needs to be frozen for next phase of planning. CEA enquired SECI on progress of new bids for RE with BESS. SECI stated that at present there is no clear visibility for such projects (RE with BESS) before 2027 as award process will take time (1-2 years). CEA mentioned that 6 GW RE potential (solar) remains untapped due to non-materialisation of BESS capacity. CTU also informed that at present no RE application are received from any of complexes in non GIB areas viz. Sanchore, Jalore Pali etc.

SECI also stated that 6GW potential at Bikaner complex can be considered for now as part of 75GW potential in Rajasthan with some portion from adjustment of potential from other Non-GIB complexes i.e. Sanchore, Jalore, Pali etc. and balance from 6GW unharnessed RE potential (with BESS) at Bikaner complex. Accordingly, it was decided that cumulatively 6 GW RE potential may be considered for planning of transmission scheme from Bikaner-IV PS.

CTU stated that considering requirement of 6 GW evacuation capacity from Bikaner-IV PS, 3GW RE capacity to be evacuated through EHVAC system as part of present proposal and balance 3GW along with Bhadla-IV potential (4GW) through HVDC system, which is under planning. In view of that system studies were carried and proposed Transmission scheme is as under

Proposed Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-3)

Bikaner-IV : 3GW (Solar)

- Establishment of 3x1500 MVA, 765/400 kV & 3x500 MVA 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVA (765kV) Bus Reactor & 2x125 MVA (420kV) Bus Reactor at a suitable location near Bikaner *
- 220kV line bays (4 nos.) for RE connectivity at Bikaner-IV PS
- STATCOM (2x±300MVA) along with MSC (4x125 MVA) & MSR (2x125 MVA) along with 2 nos. 400kV line bays at Bikaner-IV PS
- Establishment of 765/400 kV, 3x1500 MVA Hissar-II S/s along with 2x240 MVA (765kV) Bus Reactor & 2x125 MVA (420kV) Bus Reactor at a suitable location near Hissar*
- Bikaner-IV – Hissar-II 765 kV D/c line (~320 km) along with 330 MVA switchable line reactor for each circuit at each end
- Bikaner-IV – Sikar-II 400 kV D/c line (Quad) (~270 km) along with 63 MVA switchable line reactor for each circuit at each end
- Hissar-II- Jind(PG) 400 kV D/c line (Quad) (~50 km)
- Hissar-II- Patran 400 kV D/c line (Quad) (~110 km)

Study files for solar maximized scenario was shared on 13.10.23 with all NR constituents. Grid-India vide mail 18.10.23 sent their observations on proposal and studies. Same was also deliberated in meeting. Reply of major observation is under

- 1. Demands of Punjab, Haryana & Rajasthan seem to be on the lower side looking at demand observed in the current time-period.**

CTU stated that in winter scenario, demand of Punjab and Haryana is lesser, and they will review the demand and change it in revised files for 2026 time frame, however Rajasthan demand in winter and summer scenario is as per 20th EPS demand (load+losses).

2. N-1 non compliance of ICTs in Rajasthan

CTU stated that most of the 400/220kV ICTs which are N-1 non compliant are in intra state (STUs), some of which are already highlighted in OCC/NRPC meetings. Regarding ICTs at Kota (PG), high loading is observed due to no KTPS generation considered in study files. RVPN confirmed that KTPS units will be operational by 2027-28 and decision on KTPS units to take out of the system will be taken by management. In view of that in revised files KTPS unit will kept on. Loading of 400/220kV Bassi ICT is marginally higher in N-1 (320MW). Same will be reviewed and taken up separately in consultation with Grid-India & CEA based on real time loading.

3. High fault current at 220kV buses

Regarding high fault level of some of RE injection stations (at 220kV level), CTUIL stated that fault level of 220 kV level at old RE pooling stations i.e. Bhadla(PG), Bhadla-II, Fatehgarh-II is marginally higher (designed for 40kA). It was informed that during fault level calculation fault level contribution from RE generation of designated 220kV bus is not considered. Same philosophy is also considered for SCR calculation. Grid-India stated that in case of distant fault, there is a possibility that RE generation will also contribute in fault. CTU stated that matter will be separately discussed with Grid-India as new proposed scheme does not influence the fault levels on these RE pooling stations.

4. Angular difference between 765kV Bikaner-IV & Hissar

Grid-India stated that under N-2 contingency of 765kV Bikaner-IV – Hissar line (both ckts out), there are severe file convergence issues. CTUIL stated that in the event of N-2 contingency, there will be no path available for evacuation of 3GW power from Bikaner-IV PS. CTUIL stated that they have carried out various cases to provide anchoring to Bikaner-IV from nearby ISTS stations i.e. Bikaner(PG), Bikaner-II, Bikaner-III. In all scenarios, power will flow towards Bikaner-IV and evacuation system requirement will increase substantially in base case.

Based on deliberations, various other options were also carried out i.e. LILO of Bikaner-III – Neemrana-II D/c at Bikaner-IV, Bikaner-IV-Sikar-II 765kV D/c etc. CTU stated that with LILO of Bikaner-III – Neemrana-II D/c at Bikaner-IV case is converged in 'N-2' contingency but angular separation will be more than 30 degree.

In the meeting, CTUIL stated that critical loading is observed in 220kV Patran – Patran (PSTCL) D/c line in studies. PSTCL stated that the loading was higher in past and for that they have carried out the LILO of 220 kV Mansa-Sunam line at Patran (Indi Grid). PSTCL stated that in future Peddy scenarios, line loading will be higher and N-1 non compliant. CTU stated that even without injection at Patran from above scheme, 220kV Patran – Patran (PSTCL) D/c line remains N-1 non compliant and PSTCL may take suitable measures to relieve the loading. PSTCL stated that whether reconductoring of line will be done by PSTCL or it will be in interstate. CTU stated that ownership of this intra state line is with STU (PSTCL), accordingly, suitable strengthening needs to be planned and implemented by PSTCL as part of intra state scheme. PSTCL stated that they will revert on the same.

Based on Grid-India comments (S no.4), CTU stated that N-1-1 or N-2 is a rare contingency and deliberations are required on N-1-1/N-2 compliance in planning studies in reference to manual on transmission planning criteria 2023. In view of above in the first meeting it was concluded that that other options may be explored by CTU in consultation with CEA and Grid-India and will be discussed in next Joint study meeting.

Gist of discussion held in Second Joint study meeting on 23.10.23

In the 2nd joint study meeting held on 23.10.23, CTU stated that they have explored various other alternatives and in new proposal, an intermediate substation in 765/400kV Churu is created with its connectivity to LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s and Fatehabad (PG) through 400 kV D/c line. To provide anchoring at Bikaner-IV, LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line (2nd) at Bikaner-IV PS is also considered. With above revised proposal, power flow is in order and angular separation incl. in N-2 contingency and voltages are within limit. Revised Study files for solar maximized scenario was shared with all constituents on 20.10.23. **Result of system studies enclosed in Exhibit-I**

Grid-India stated that the proposed system is optimal & balanced and loading is equally distributed among various feeders. The proposed system is stable and comply the N-1 & N-1-1 requirements. CEA stated that as per transmission planning criteria 2023, under N-1-1, some of the equipment may be loaded up to their emergency limits. To bring the system parameters back within their normal limits, load shedding/re-scheduling of generation may have to be done, either manually or through automatic system protection schemes (SPS).

In view of that in planning stage, N-1-1 or N-2 criteria may not be considered except in critical lines (Inter regional corridors) as it will increase the transmission system requirement. Therefore, in revised proposal a direct interconnection between Bikaner-IV and Siwani may be considered and 400kV interconnection towards Fatehabad may be planned from Siwani in place of via Churu.

CTU stated that in above alternative with direct interconnection to Siwani, file is converged, and power flow is in order, however angular separation between Siwani and Bikaner-IV is more than 50 degree in N-2 contingency in case of direct interconnection of Bikaner-IV and Siwani. The angular separation would be more than 30 degree even with significant less RE capacity (<1GW) at Bikaner-IV PS.

Grid-India stated that it is not recommended that SPS implementation is considered at planning stage . SPS requirement will generally come when study assumption considered in studies during planning may deviate at later stage i.e. load change or delay in interlinked transmission system which influence the load flow.

Further, in case of direct interconnection of Bikaner-IV to Siwani i.e. not considering 765/400kV Churu in between and onward transmission system, the Transmission system is kind of radial system connected with RE generation pocket and poses stability issues in various operational scenarios in future. Grid-India also emphasised that some margin should be kept in planning studies for operational scenarios. Further, the angular difference (>30 degrees) in N-1-1 /N-2 may cause problems in synchronization of lines after corridor outage. The proposed system is stable and well interconnected with Grid provides reliable power evacuation under various operational scenarios

CEA stated that N-1-1 or N-2 is a rare contingency and deliberations are required on N-1-1/N-2 compliance in planning studies as it may incur additional investment for strengthening of transmission system.

CTU stated that in above case the transmission scheme i.e. establishment of Churu substation along with LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s and Churu – Fatehabad (PG) 400 kV D/c line will improve the system resiliency and address the Grid India concern for synchronization of lines after corridor outage in N-1-1 /N-2) contingency

Considering 3.6GW generation at Bikaner-IV PS, loading and angular separation in contingency is as under:

Transmission Line	N-1 (Loading)	N-1-1 (N-2) (Degree)
765 kV Bikaner-IV-Churu D/c line	14.1 (3098 MW)	29.2
765 kV Churu- Siwani D/c line	7.7 (2743 MW)	28.6

CTU further stated that in next phase, EHVAC/HVDC system (5-6GW) for evaluation of RE power from Bhadla-IV potential & balance potential of Bikaner-IV (~2.4GW) towards UP/outside NR region is under planning and will be taken up in subsequent meetings. CTU requested CEA to convene a joint meeting for compliance of N-1-1/N-2 in planning studies in reference of planning criteria 2023 by next week. CEA agreed for same. PSTCL stated that they will take suitable measures to relieve loading of 220kV Patran – Patran (PSTCL) D/c line in matching timeframe of above agreed ISTS scheme.

In view of above deliberations, it was decided that CEA will convene a joint meeting with CTU and Grid-India for deliberation on compliance of N-1-1/N-2 compliance in planning studies. Based on outcome of above CEA meeting, proposed system requirement will be reviewed and if system requirement is reduced (with scheme like Bikaner-IV – Siwani direct interconnection), same shall be suitably incorporated as part of minutes of CMETS-NR meeting.

Considering grant of connectivity to new RE generators in Bikaner complex as well as for evacuation of power beyond Bikaner complex, following transmission scheme was agreed in Joint study meeting for evacuation of power from Rajasthan REZ Ph-IV (Part-3:3.6GW) [Bikaner complex] and to be taken up in CMETS-NR meeting for finalization.

Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-3 :3.6GW)

Bikaner-IV: 3.6GW (Solar)

- Establishment of 4x1500 MVA, 765/400 kV & 4x500 MVA, 400/220 kV Bikaner-IV Pooling Station along with 2x240 MVar (765kV) & 2x125 MVar (420kV) Bus Reactors at a suitable location near Bikaner *

Future provisions at Bikaner-IV PS:

Space for

- 765/400kV ICT along with bays- 2 no.
- 765 kV line bays along with switchable line reactors – 10 nos.
- 765kV Bus Reactor along with bay: 1 no.
- 400 kV line bays along with switchable line reactor –4 nos.

- 400 kV line bays—4 nos.
- 400/220kV ICT along with bays -6 nos.
- 400 kV Bus Reactor along with bay: 1 no.
- 400kV Sectionalization bay: 2 sets
- 220 kV line bays for connectivity of RE Applications -11 nos.
- 220kV Sectionalization bay: 3 sets
- 220 kV BC (2 nos.) and 220 kV TBC (2 nos.)
- *along with provision of 80MVar and 110MVar spare reactor (Single phase)**

- 220kV line bays (5 nos.) for RE connectivity at Bikaner-IV PS
- 400kv line bay (1 no.) for RE connectivity at Bikaner-IV PS
- 220kV Sectionalization bay (1 set) along with BC (2 nos.) and 220 kV TBC (2 nos.) at Bikaner-IV PS
- 400kV Sectionalization bay (1 set) at Bikaner-IV PS
- STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar) along with 2 nos. 400kV line bays at Bikaner-IV PS
- LILO of one ckt of 765 kV Bikaner-III -Neemrana-II D/c line (2nd) at Bikaner-IV PS (~20 km) along with 330 MVAR switchable line reactor at Bikaner-IV PS end of 765 kV Bikaner-IV - Neemrana-II line (formed after LILO)
- Establishment of 765/400 kV, 2x1500 MVA S/s at suitable location near Churu along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor

Future provisions at Churu S/s:

Space for

- 765/400kV ICTs along with bays- 4
- 765 kV line bays along with switchable line reactors – 12
- 765kV Bus Reactor along with bay: 1 nos.
- 400 kV line bays along with switchable line reactor –8
- 400 kV Bus Reactor along with bays: 1 no.
- 400kV Sectionalization bay: 2 sets
- 400/220kV ICT along with bays -4 nos. **
- 220 kV line bays for drawl -4 nos. **
- 220kV Sectionalization bay: 2 sets **
- *along with provision of 80MVar spare reactor (Single phase)**
- **Drawl requirement at Churu to be confirmed by RVPN**

- Bikaner-IV PS – Churu 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at Bikaner-IV PS end (~175 km)
- LILO of one ckt of 765 kV Sikar-II (PG) -Khetri (PG) D/c line at Churu S/s (~80 km)
- Churu – Fatehabad (PG) 400 kV D/c line (Quad) along with 80 MVar switchable line reactor for each circuit at Churu S/s end (~165 km)
- Establishment of 765/400 kV, 3x1500 MVA S/s at suitable location near Siwani (Distt. Bhiwani) along with 2x240 MVar (765kV) Bus Reactor & 2x125 MVar (420kV) Bus Reactor*

Future provisions at Siwani S/s:

Space for

- 765/400kV ICT along with bays- 3
- 765 kV line bays along with switchable line reactors – 6
- 765kV Bus Reactor along with bay: 1 nos.
- 400 kV line bays along with switchable line reactor –10

- Siwani – Churu 765 kV D/c line (~110 km)
- Siwani – Jind (PG) 400 kV D/c line (Quad Moose) (~100 km)
- Siwani – Patran (Indi Grid) 400 kV D/c line (Quad) (~150 km) along with 63 MVar switchable line reactor for each circuit at Siwani S/s end

Tentative commissioning schedule : 24 months from allocation

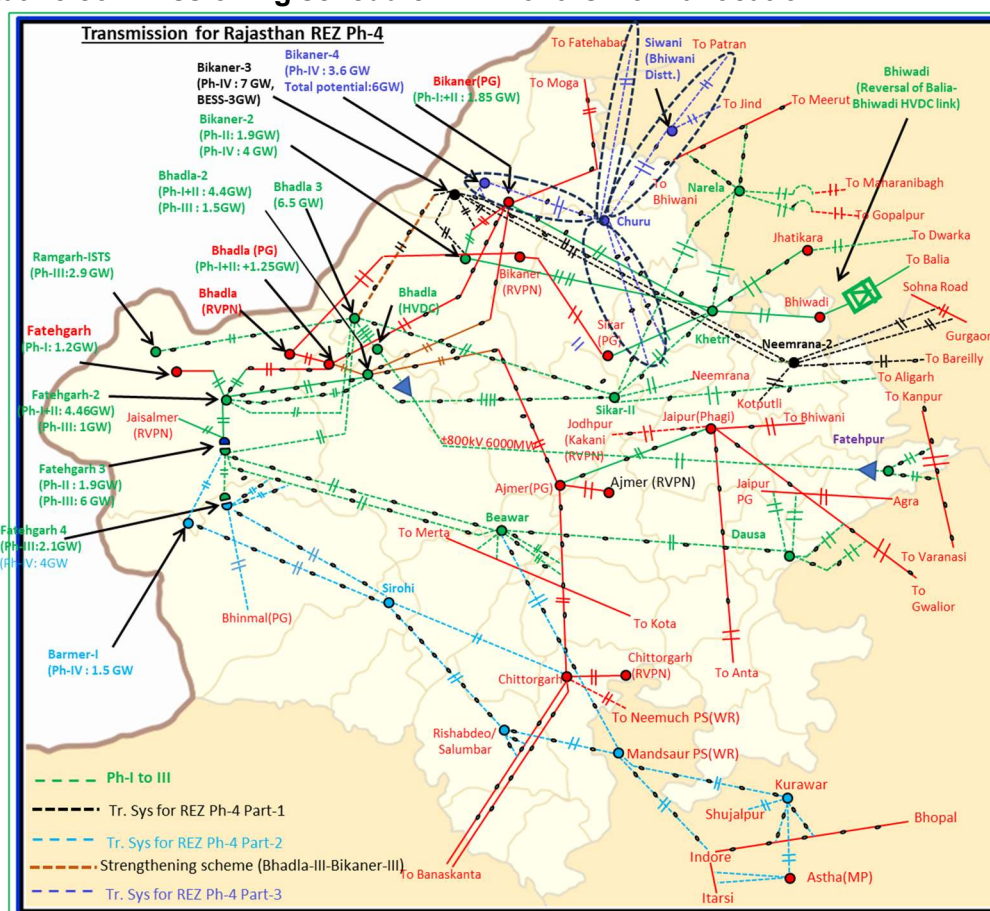
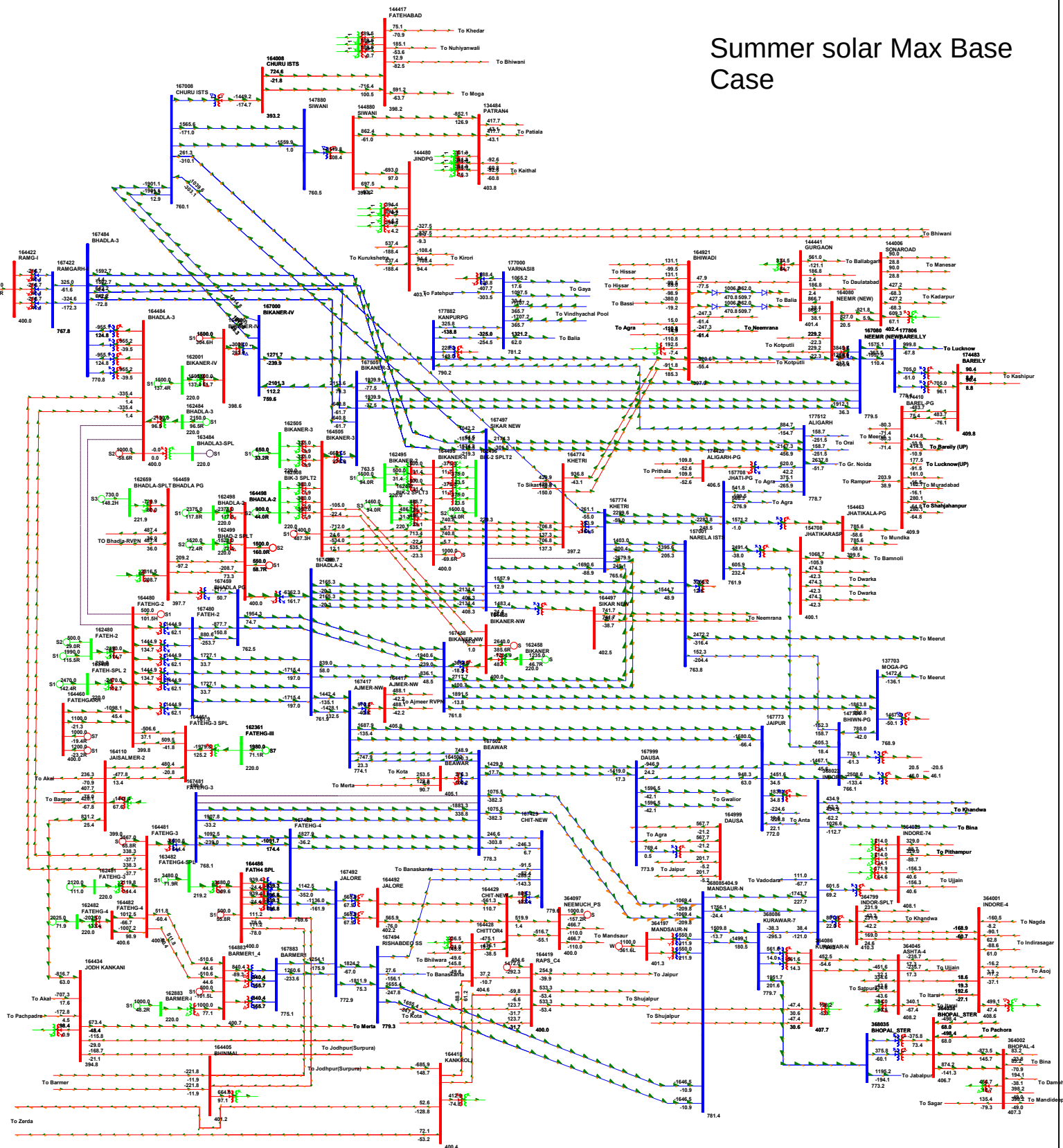


Fig 1: Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-3 :3.6GW) (Bikaner Complex)

Summer solar Max Base Case



Winter Solar Max Base Case

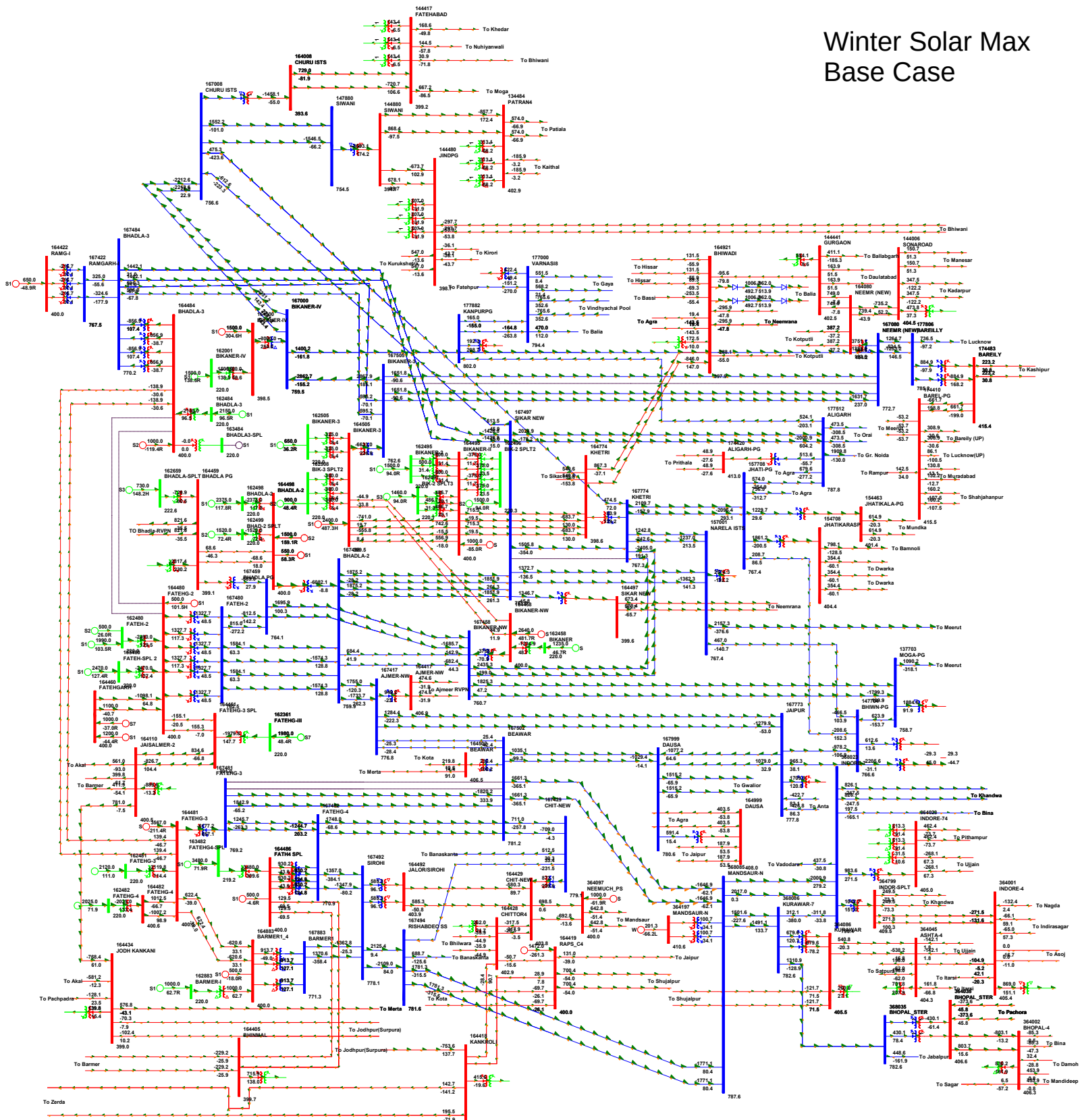
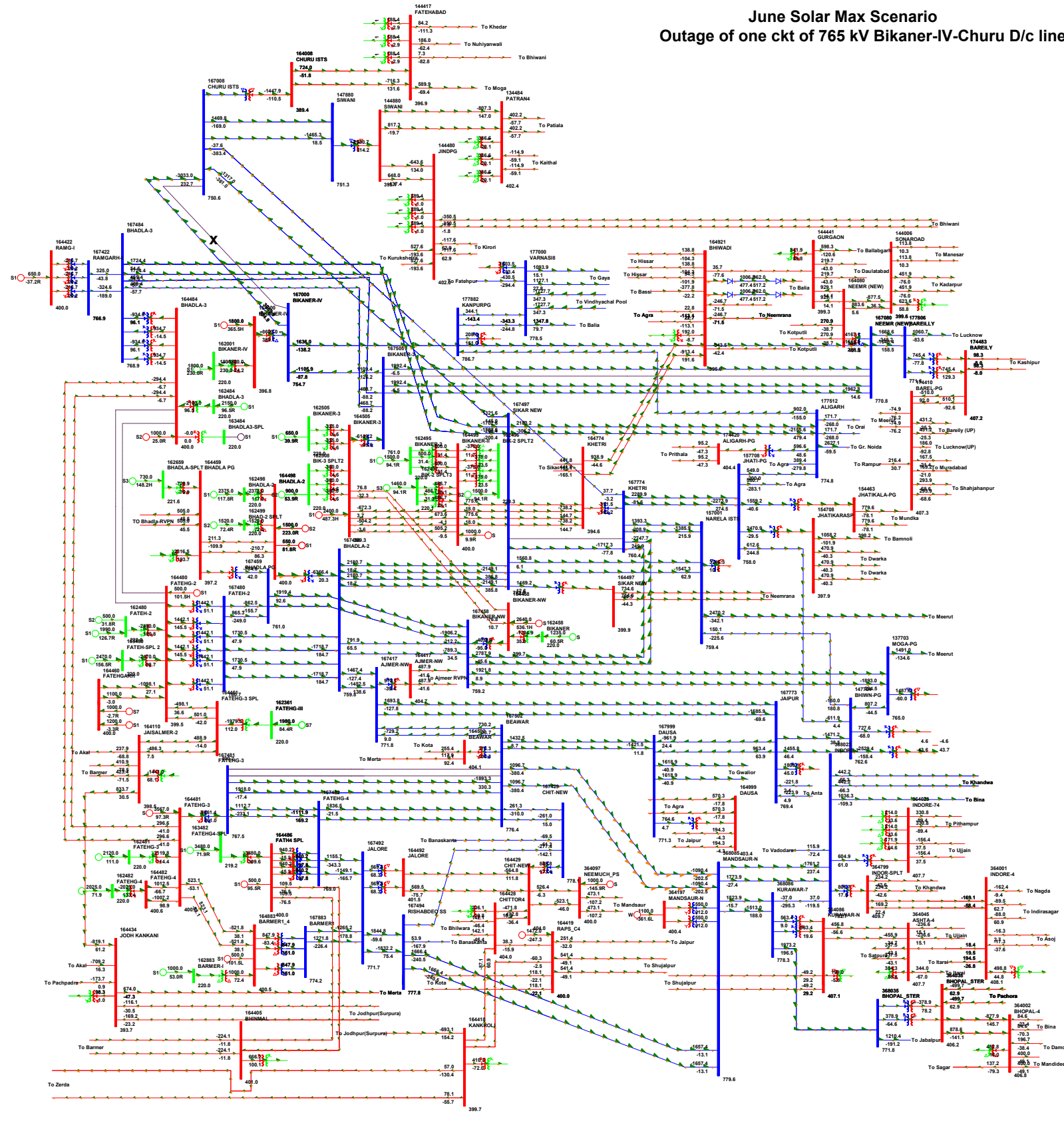


Exhibit-I



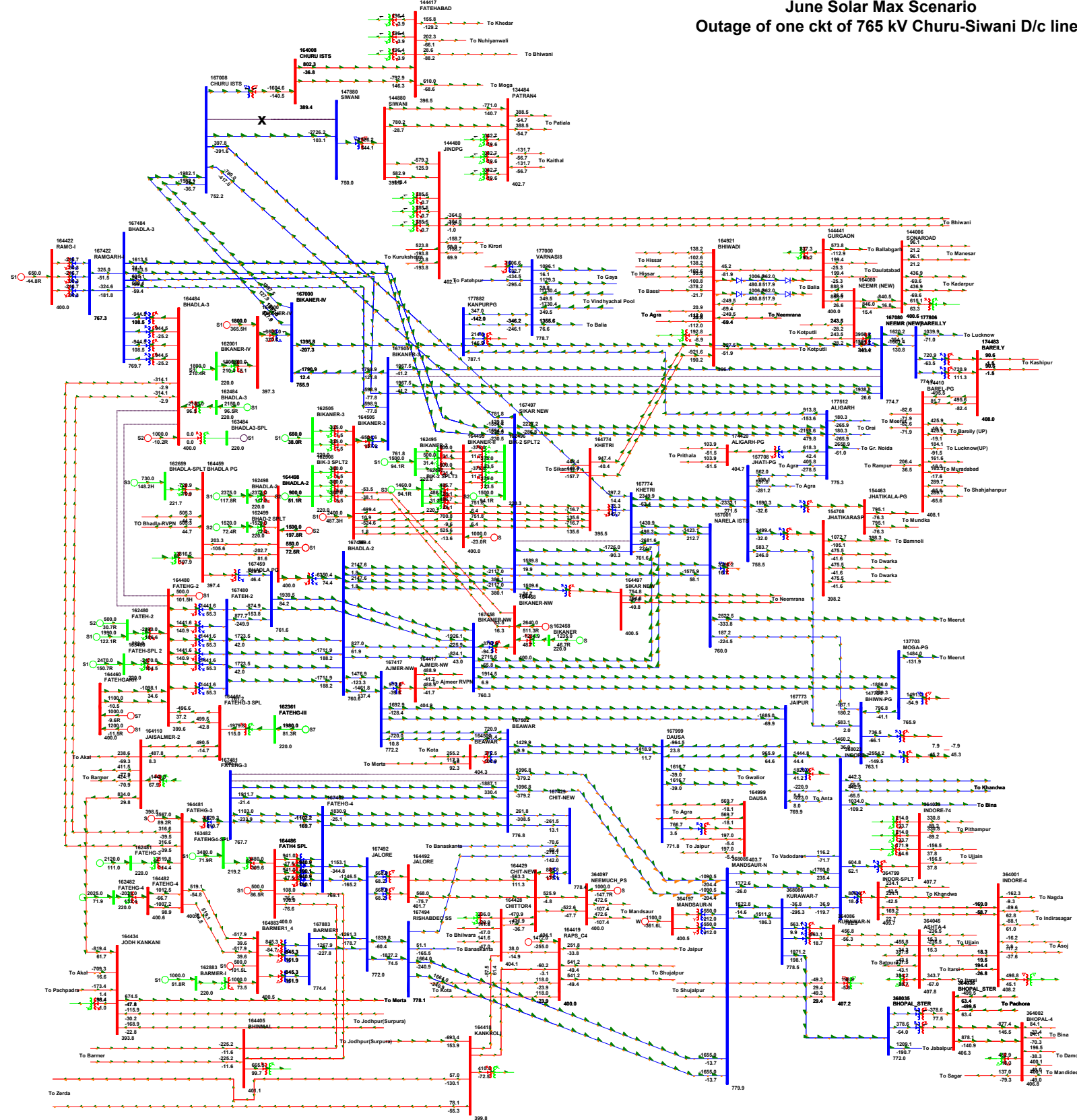
Transmission System for Rajasthan REZ Phase-IV (Part-3: 3.6 GW) (Bikaner Complex)

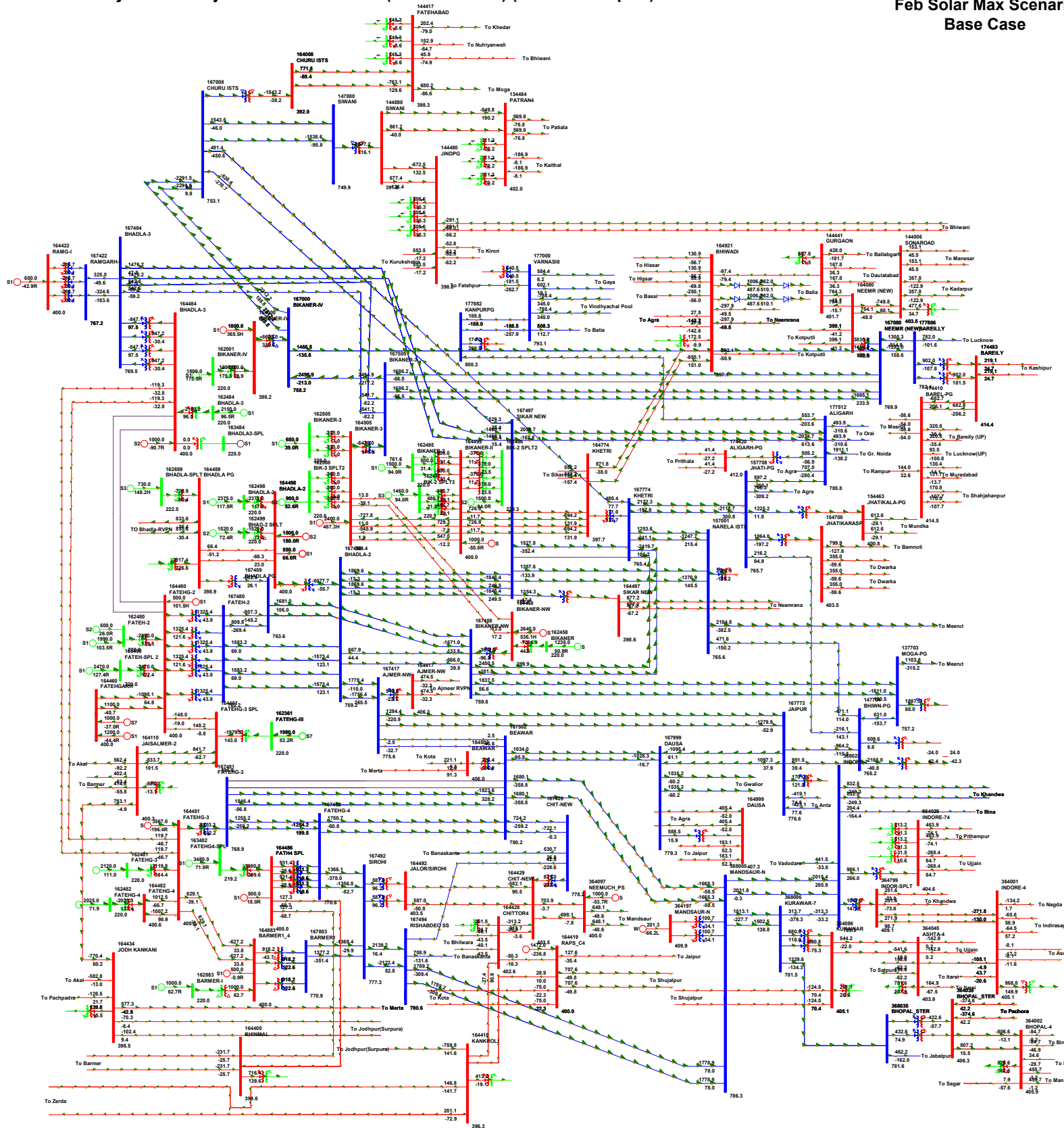
June Solar Max Scenario
Outage of one ckt of 765 kV Bikaner-IV-Churu D/c line



Transmission System for Rajasthan REZ Phase-IV (Part-3: 3.6 GW) (Bikaner Complex)

June Solar Max Scenario
Outage of one ckt of 765 kV Churu-Siwani D/c line



Feb Solar Max Scenario
Base Case

Transmission System for Rajasthan REZ Phase-IV (Part-3: 3.6 GW) (Bikaner Complex)

Feb Solar Max Scenario
Outage of one ckt of 400 kV RAPS-Sujalpur D/c line

